



ALGORITHMIC BIAS AND FEEDBACK: INSTABILITIES and DYNAMICS

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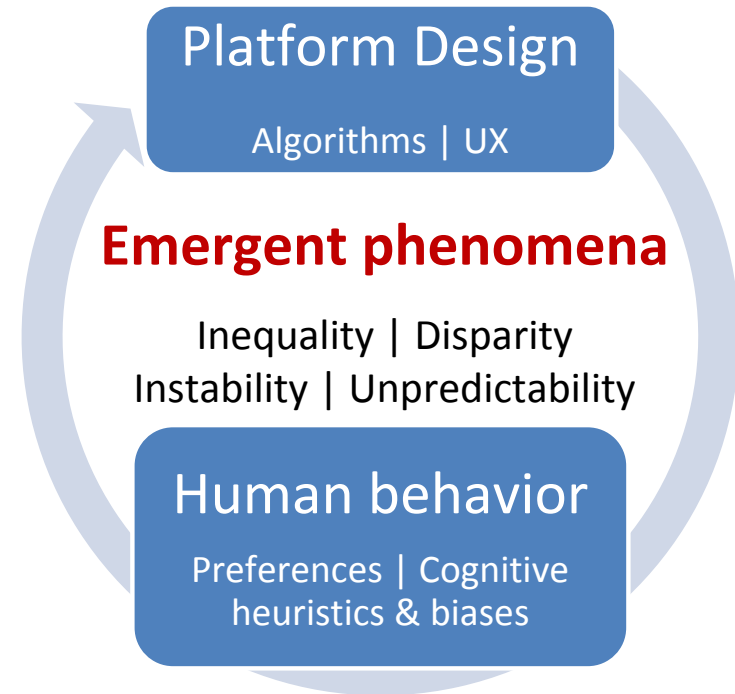
Spring 2025

DSCI 531



Dynamics of algorithms

- The intricate **interplay** between **algorithms** and **people** within online platforms leads to **emergent, unpredictable** behaviors
- **Unstable** systems: similar initial conditions result in vastly different outcomes
- **Biased** systems create inequities and disparities





Recommender systems

Recommender systems mediate access to information, shape how content is discovered, connections formed, and decisions are made.

- Crowdsourced rankings aggregate many individual decisions to identify best options
 - Best seller lists
 - College rankings
- Ranking online
 - Reddit: most interesting news
 - StackOverflow: best answers to questions
 - Twitter: trending topics
- Recommendation algorithms
 - Personalize content





Emergent behavior in recommender systems

- **Algorithmic instabilities and structural disparities**
 - Algorithms interact with existing human biases.
 - Feedback loops emerge, systematically disadvantage target groups over time.
 - Cumulative advantage: biases compound over time
 - Structural disparities can emerge, e.g., cumulative disadvantage for minority groups
 - Algorithms can either reinforce systemic biases or counteract them
- **Types of algorithms and behaviors**
 - Instabilities in **ranking** (e.g., aggregate decisions of crowds)
 - MusicLab study: Inequality and unpredictability of popularity
 - Wisdom of crowds (e.g., Stack Overflow): do best items float to the top?
 - Can be explored in simulations
 - Instabilities in **personalization**
 - Personalized recommendations (e.g., Twitter feed): inequality of outcomes
 - Dynamic personalization: algorithmic chaos
 - **Disparities** in growing networks
 - Gender disparity emerges in growing citation networks



INSTABILITIES IN RANKING



Algorithmic + Cognitive bias

Home shows you top Tweets first

See latest Tweets instead
You'll be switched back Home after you've been away for a while.

[View content preferences](#)

Trends for you

- Politics · Trending
ISIS
26.6K Tweets
- Trending in Chile
Pichilemu
- Trending in Chile
#NoRobenMisAhorros
6,820 Tweets
- Politics · Trending
Biden
877K Tweets
- Trending in Chile
#Mañalich
3,087 Tweets

Who to follow

- neil_ferguson** @neil_ferguson [Follow](#)
- Bill Kristol** @BillKristol [Follow](#)
- loreana penjean** @lorepjen [Follow](#)

Toby Walsh @TobyWalsh · 4m
BBC News - AI cannot be recognised as an inventor, US rules

BBC NEWS

AI cannot be recognised as an inventor, US rules
The US Patent and Trademark Office rules only "natural persons" can legally be inventors.
bbc.co.uk

Location, location, location...

Where information is presented affects how much attention it receives

- Ranking bias in search
 - People click on the first few links in search results
- e.g., Twitter shows Personalized tweets or Latest tweets (left)
- Ranking by popularity vs recency affects success in peer recommendation [Lerman & Hogg, 2014].



Popularity ranking

- **Crowdsourced ranking**

- aggregates decisions of many people to rank items
- “quality items float to the top”

- **Instability in popularity ranking**

- Music Lab study [Salganik+ 2006]
- Few items become hits; most languish in obscurity
- Can’t predict what items become hits
 - Harry Potter book was rejected 9 times

Popularity ranking: Popular way to organize content on Twitter, Reddit, StackOverflow, ...

The screenshot shows a web interface for crowdsourced ranking. At the top, there is a red 'Finish' button and a prompt: 'Please recommend science topics from the list below'. Below this, there is a list of seven science topics, each with a thumbs up/down icon and a 'recommend' button. The topics are:

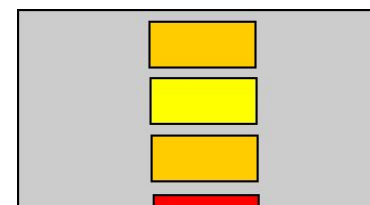
- Grab the Leash: Dog Walkers More Likely to Reach Exercise Benchmarks**
Man's best friend may provide more than just faithful companionship: A new study led by a Michigan State University researcher shows people who owned and walked their dogs were 34 percent more likely to meet federal benchmarks on physical activity.
- No Two of Us Are Alike -- Even Identical Twins: Pinpointing Genetic Determinants of Schizophrenia**
Just like snowflakes, no two people are alike, even if they're identical twins according to new genetic research from The University of Western Ontario. Molecular geneticist Shiva Singh has been working with psychiatrist Dr. Richard O'Reilly to determine the genetic sequencing of schizophrenia using...
- Lab-Grown Meat Would 'Cut Emissions and Save Energy'**
Meat grown using tissue engineering techniques, so-called 'cultured meat', would generate up to 96% lower greenhouse gas emissions than conventionally produced meat, according to a new study.
- Still Counting Calories? Your Weight-Loss Plan May Be Outdated**
The most detailed long-term analysis of the factors that influence weight gain shows that conventional wisdom may not be the best approach.
- Happy Guys Finish Last, Says New Study On Sexual Attractiveness**
"Women find happy guys significantly less sexually attractive than swaggering or brooding men, according to a new University of British Columbia study that helps to explain the enduring allure of ""bad boys"" and other iconic gender types."
- Storing Water for a Dry Day Leads to Suits**
A small water utility in California sued to challenge the wealthy farming interests that dominate two of the country's largest water banks.
- Governor Says Montana Was Misled on Oil Spill**



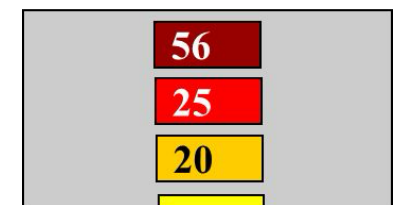
MusicLab study

Matthew J. Salganik, Peter Sheridan Dodds, Duncan J. Watts. Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market. *Science* 10 Feb 2006 : 854-856

- Online experiment to simulate a cultural market
- Users shown a list of songs in one of two conditions
 - **Independent:** songs in random order
 - **Social influence:** songs ranked by popularity
- Users download songs they like
 - Popularity = fraction of all downloads a song receives
 - Quality = popularity in the independent condition



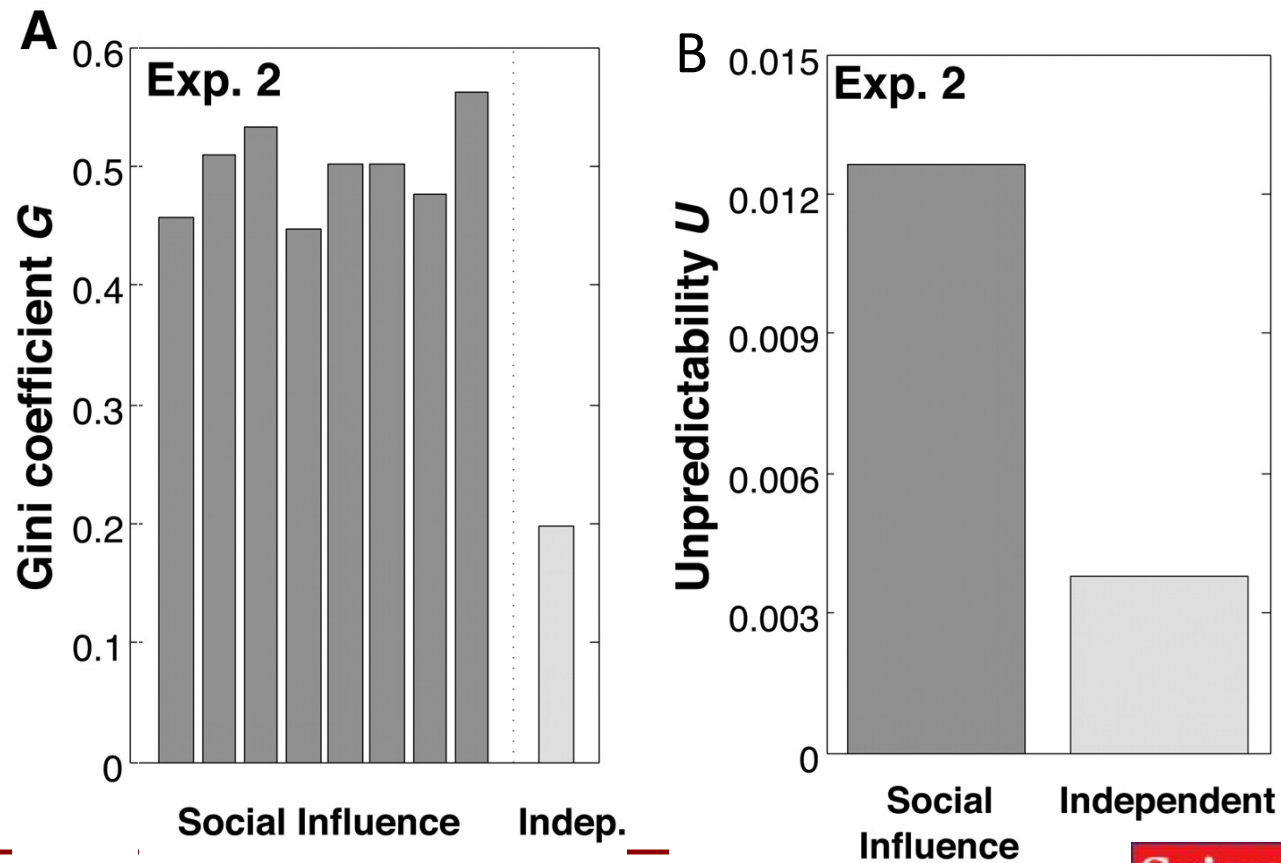
Independent



Social influence

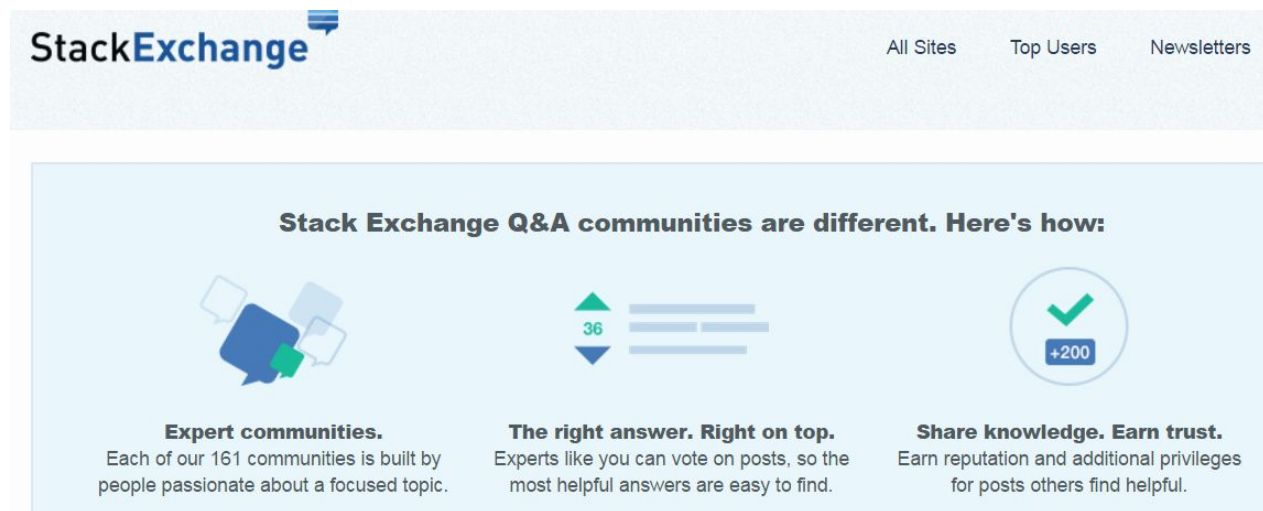


Inequality (A) and unpredictability (B) of song market share in the social influence vs independent conditions worlds





COGNITIVE BIASES IN STACK EXCHANGE



[Burghardt, Alsina, Girvan, Rand, and Lerman (2017). The myopia of crowds: Cognitive load and collective evaluation of answers on stack exchange. PLOS ONE, 12(3):e0173610+.]



Anatomy of Stack Exchange

Question

Cognitive load
Number of
answers to the
question

Answers

▲ 52
▼
★
7

What's the correct way to write a `for-in` loop in JavaScript? The browser doesn't issue a complaint about either of the two approaches I show here. First, there is this approach where the iteration variable `x` is explicitly declared:

```
for (var x in set) {  
  ...  
}
```

And alternatively this approach which reads more naturally but doesn't seem correct to me:

```
for (x in set) {  
  ...  
}
```

javascript syntax for-in-loop

share improve this question

edited 5 mins ago
DavidRR
3,151 3 15 33

asked Apr 19 '11 at 13:28
futlib
2,016 4 24 43

add a comment

8 Answers

active oldest votes

▲ 48
▼
✓

Use `var`, it reduces the scope of the variable otherwise the variable looks up to the nearest closure searching for a `var` statement. If it cannot find a `var` then it is global (if you are in a strict mode, using `strict`, global variables throw an error). This can lead to problems like the following.

```
function f () {  
  for (i=0; i<5; i++);  
}  
var i = 2;  
f ();  
alert (i); // i == 5. i should be 2
```

If you write `var i` in the for loop the alert shows `2`.

JavaScript Scoping and Hoisting

share improve this answer

edited Jul 25 '13 at 7:50

answered Apr 19 '11 at 13:36
Gabriel Llamas
7,063 7 45 83

Answer features

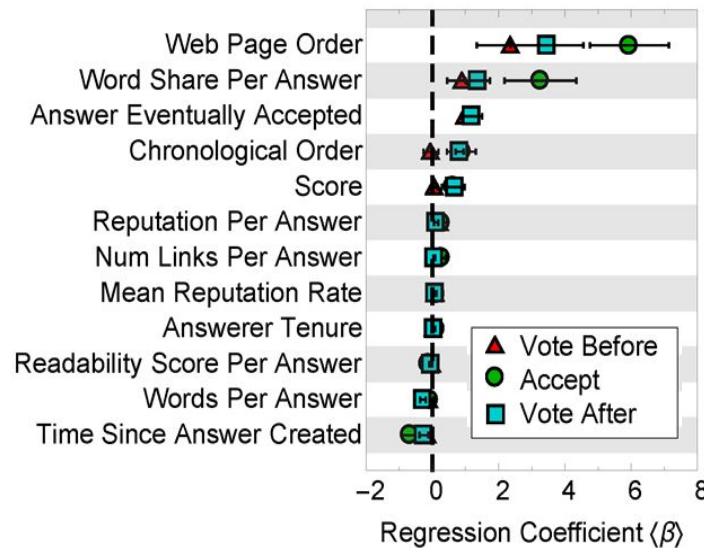
- votes/score
- accepted?
- web page order
- chrono order

- num words
- word share
- hyperlinks
- readability
- age

- answerer reputation
- tenure



Statistical analysis: regression coefficients



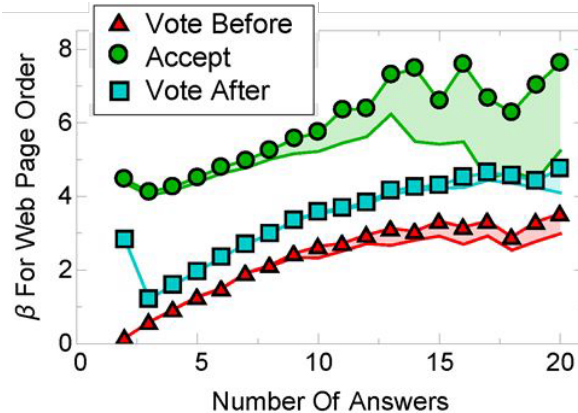
• **Rather than evaluate all answers, people use simple heuristics to choose answers to vote for or accept.** Largest coefficients are:

- Web page order $\rightarrow$ answer's position (*cf* position bias)
- Word share $\rightarrow$ fraction of the screen it occupies (*cf* availability bias)
- Answer acceptance $\rightarrow$ social proof (*cf* social influence bias)

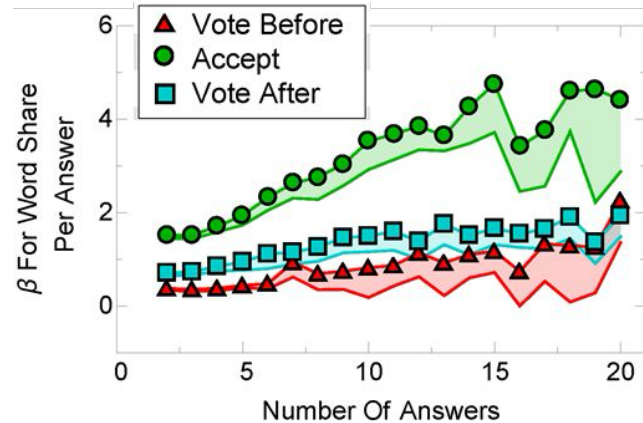


Cognitive load increases cognitive biases

Regression coefficient for web page order vs cognitive load*



Regression coefficient for word share vs cognitive load*

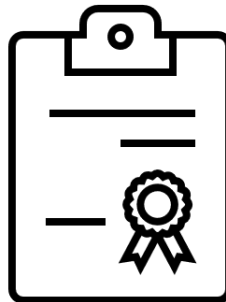


* using number of answers available to a question as a proxy of cognitive load

Cognitive load (e.g., amount of information) amplifies cognitive biases, increasing reliance on cognitive heuristics

[Burghardt et al (2017). The myopia of crowds: Cognitive load and collective evaluation of answers on stack exchange. PLOS ONE]

Does item quality matter?





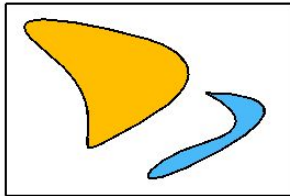
Does item quality matter?

Burghardt et al. (2020) “Origins of algorithmic instabilities in crowdsourced rankings”, in CSCW.

Online experiment:

Guess Condition

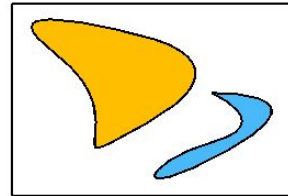
Guess the value:



Area ratio of larger
and smaller shapes

Control Condition

Choose the best value:

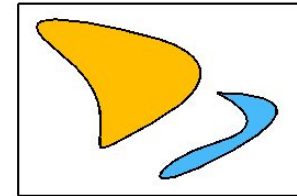


Area ratio of larger
and smaller shapes

- 2.1
- 4.6

Social Influence Condition

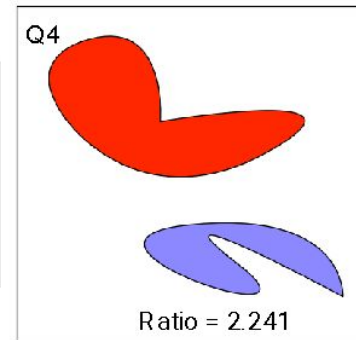
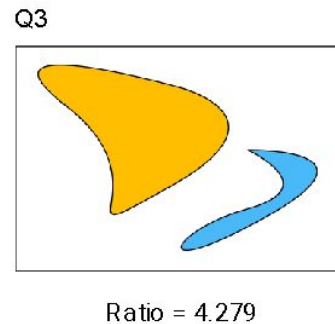
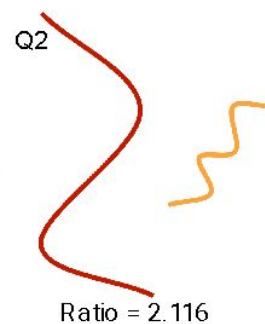
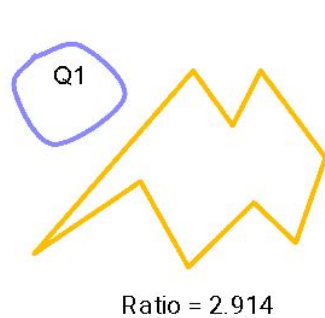
Choose the best value:



Area ratio of larger
and smaller shapes

- Most popular:* ■ 2.1
Least popular: ■ 4.6

- ~1800 Amazon Mechanical Turk workers
- **10 simple questions but hard answers**
- Toy system, but allows for isolating effects of answer quality & biases

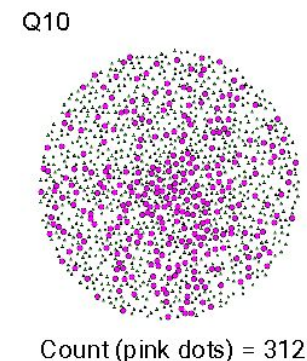
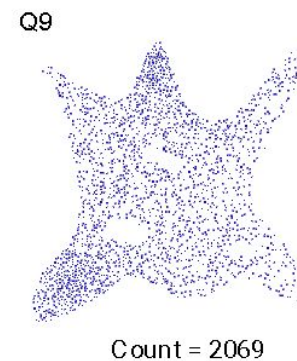
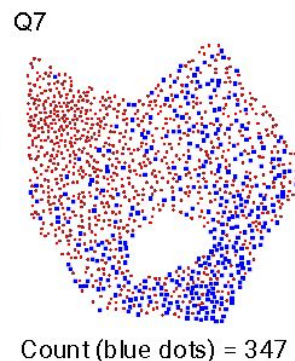
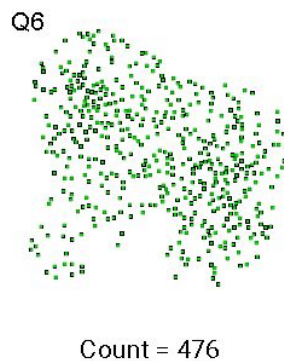


Q5 score and seven years ago our fathers brought forth on
continent, a new nation, conceived in Liberty, and
dedicated to the proposition that all men are created equal.

Now we are engaged in a great civil war, testing whether that
nation, or any nation so conceived and so dedicated, can long
endure. We are met on a great battle-field of that war. We have
come to dedicate a portion of that field, as a final resting place
for those who here gave their lives that that nation might live. It
is altogether fitting and proper that we should do this.

But, in a larger sense, we can not dedicate -- we can not
consecrate -- we can not hallow -- this ground. The brave men,
living and dead, who struggled here, have consecrated it, far
above our poor power to add or detract. The world will little
note, nor long remember what we say here, but it can never
forget what they did here. It is for us the living, rather, to be
dedicated here to the unfinished work which they who fought
here have thus far so nobly advanced. It is rather for us to be
here dedicated to the great task remaining before us -- that
from these honored dead we take increased devotion to that
cause for which they gave the last full measure of devotion --
that we here highly resolve that these dead shall not have died
in vain -- that this nation, under God, shall have a new birth of
freedom -- and that the people, by the people, for the people, shall not perish from the earth.

r's = 47

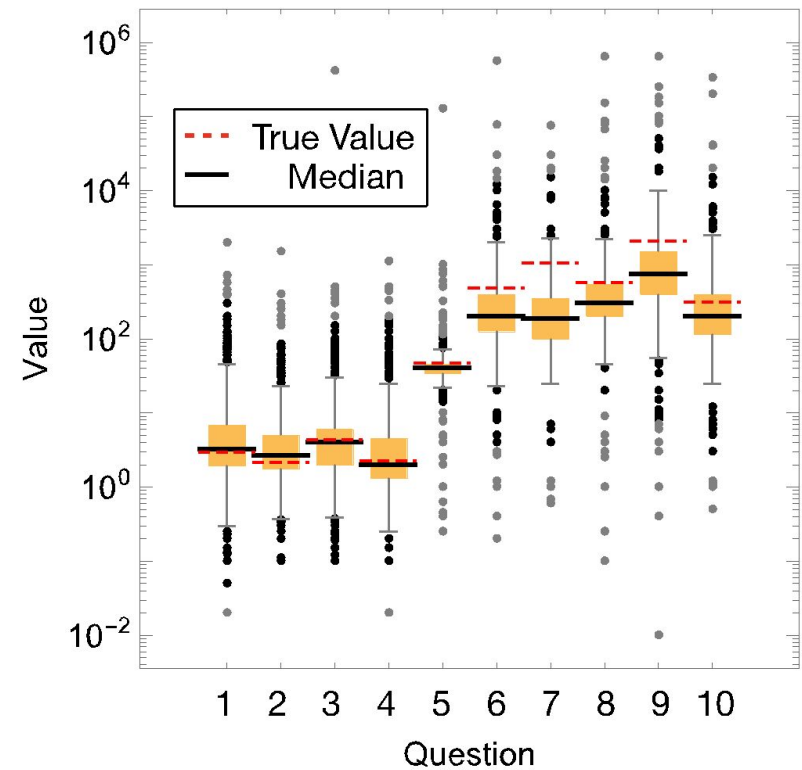




Are guesses centered around the right value?

We separately test how guesses are made

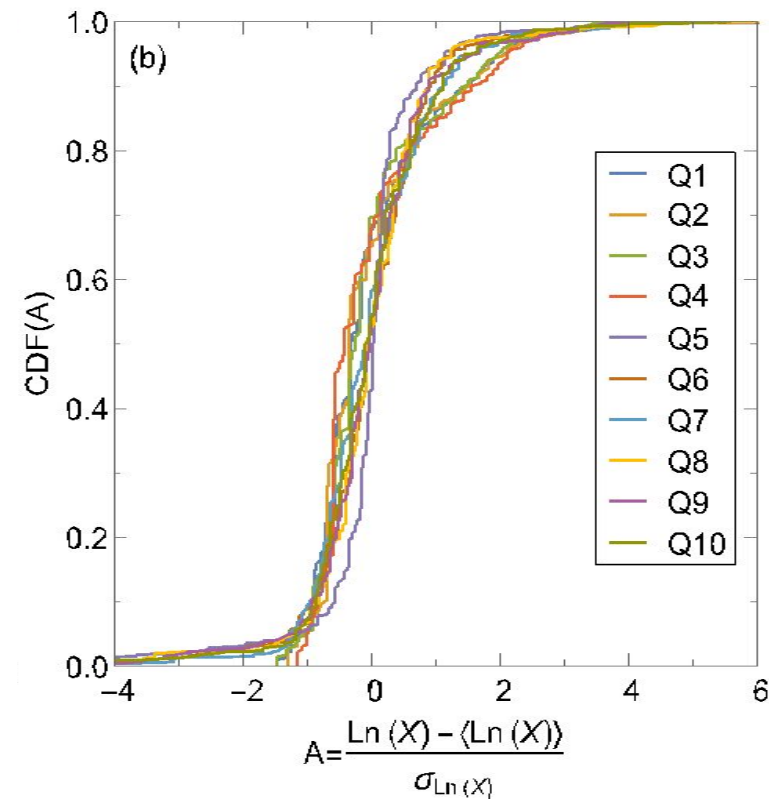
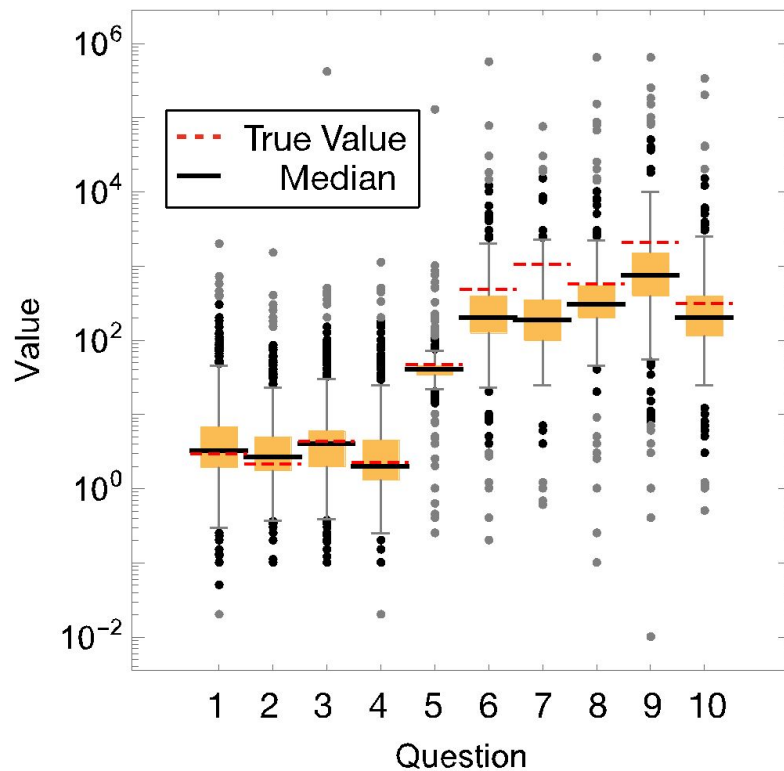
- Median guesses are centered around true value (with a few exceptions)
- Wisdom of crowds (almost) works





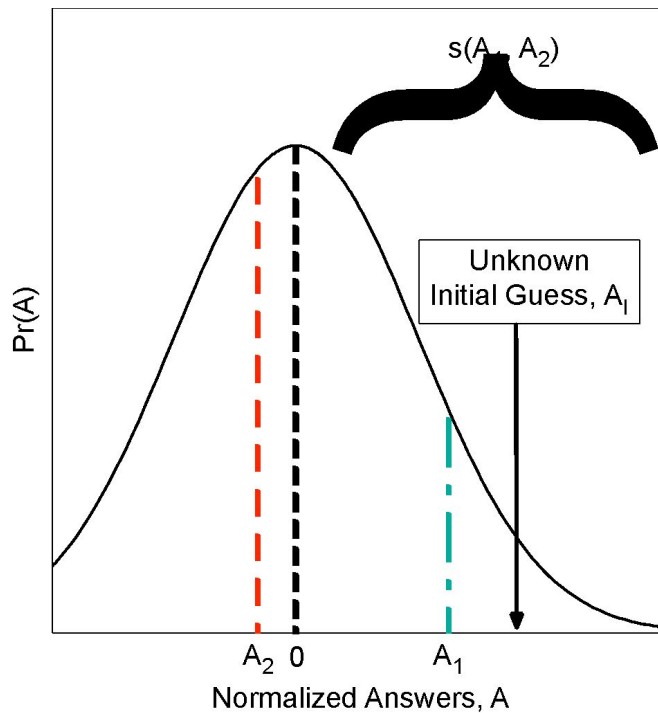
Guesses vs correct answers

Log-normal distributed guesses collapse after normalization





Model of how people choose the best answer



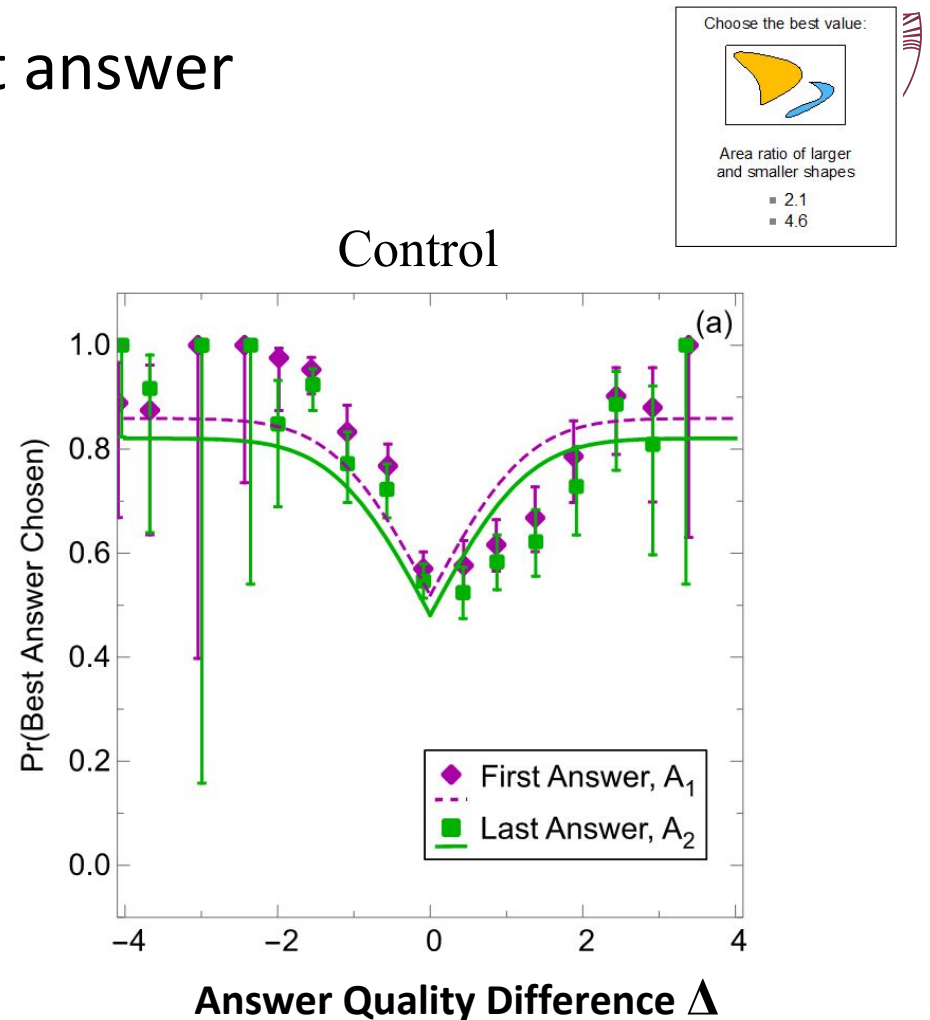
Stochastic model of decisions

- r : guess answers at random
- $(1-r)p$: choose top answer
- **Otherwise:** Choose the answer closest to your initial guess
 - Higher probability for answer closer to the correct answer
 - This models how diligent people can still get the wrong answer
- Estimate model parameters from experimental condition 1

Choosing the best answer

How does the decision to choose an answer depend on its quality?

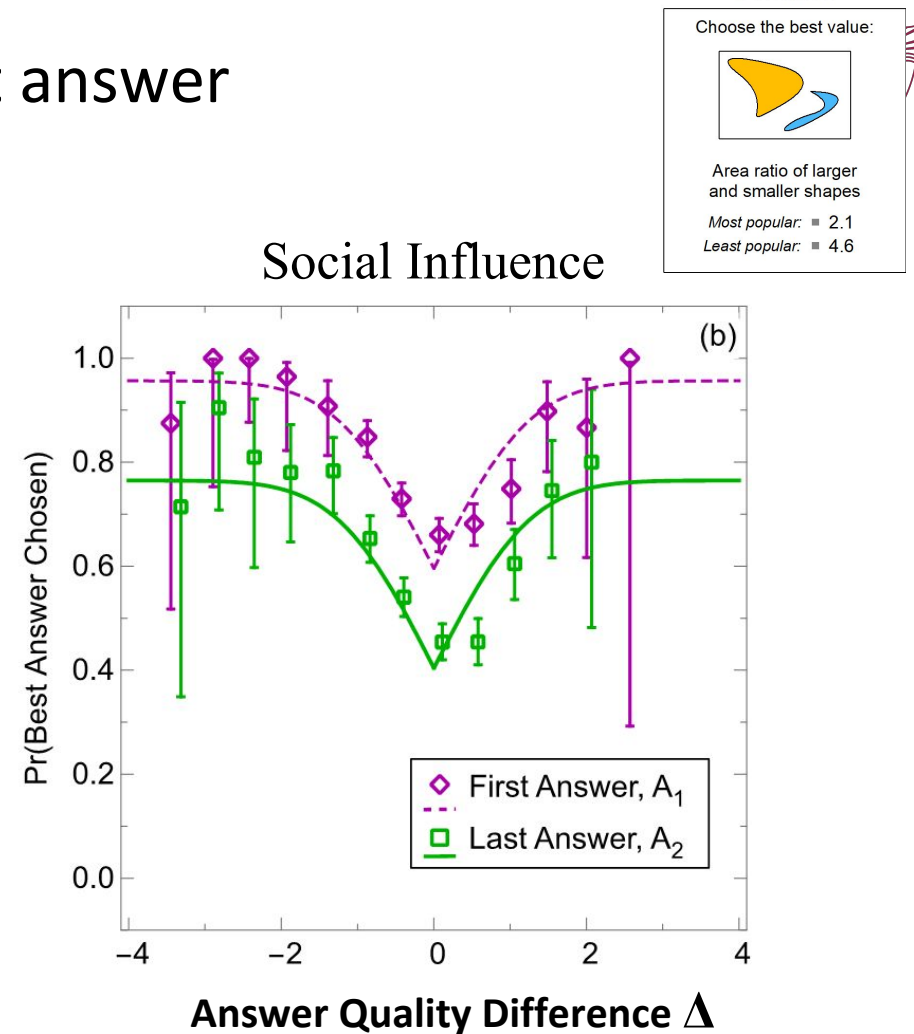
- Quality difference $\Delta = 0$: both answers equally good
- $|\Delta| > 0$: best answer is more obvious
- **Control condition results:**
 - $|\Delta| \cong 0$: Equal likelihood to pick each answer
 - Little effect of answer order



Choosing the best answer

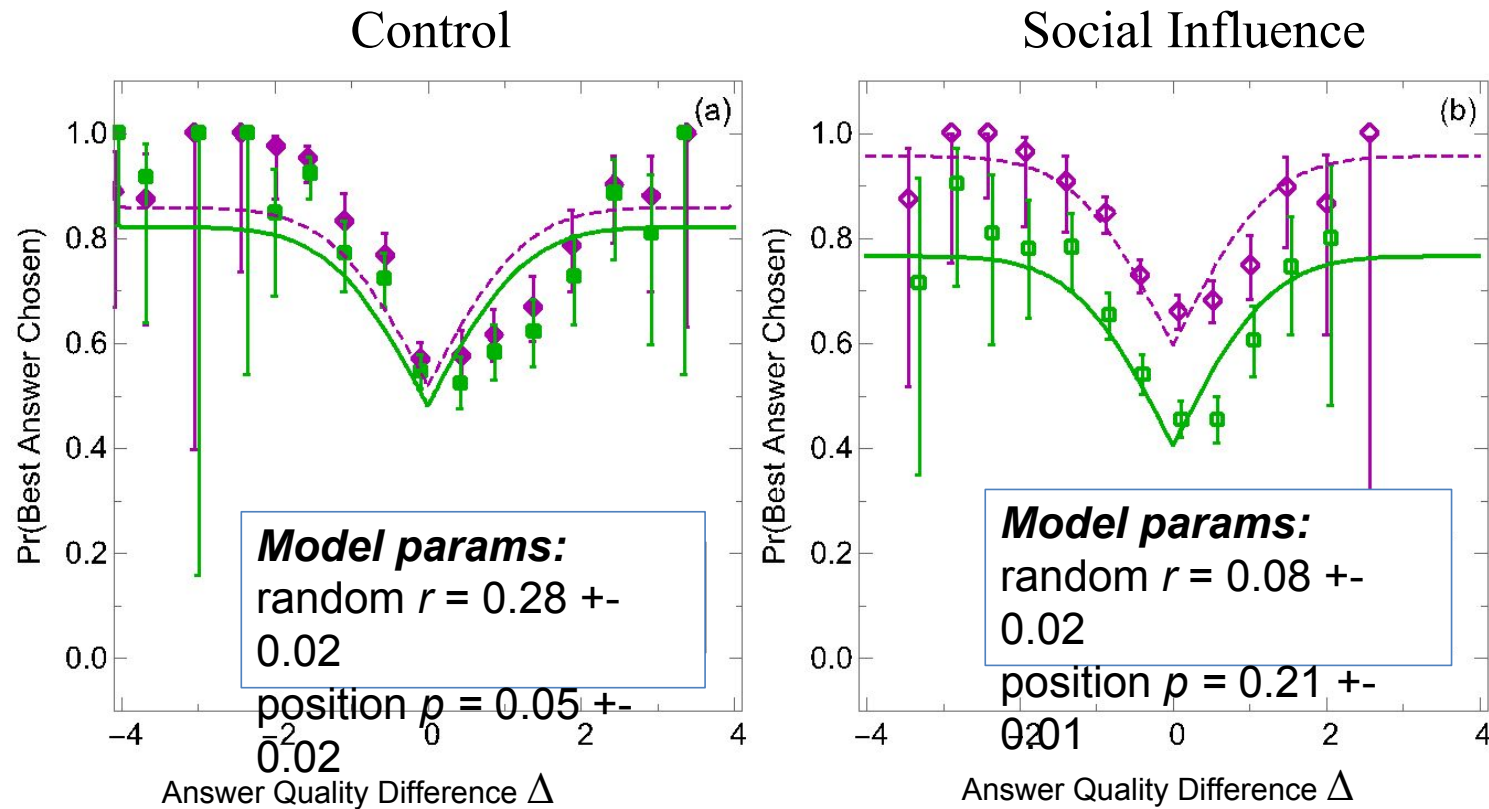
How does the decision to choose an answer depend on its quality?

- Quality difference $\Delta = 0$: both answers equally good
- $|\Delta| > 0$: best answer is more obvious
- **Social influence condition results:**
 - Same qualitative behavior across conditions
 - Top answer preferred





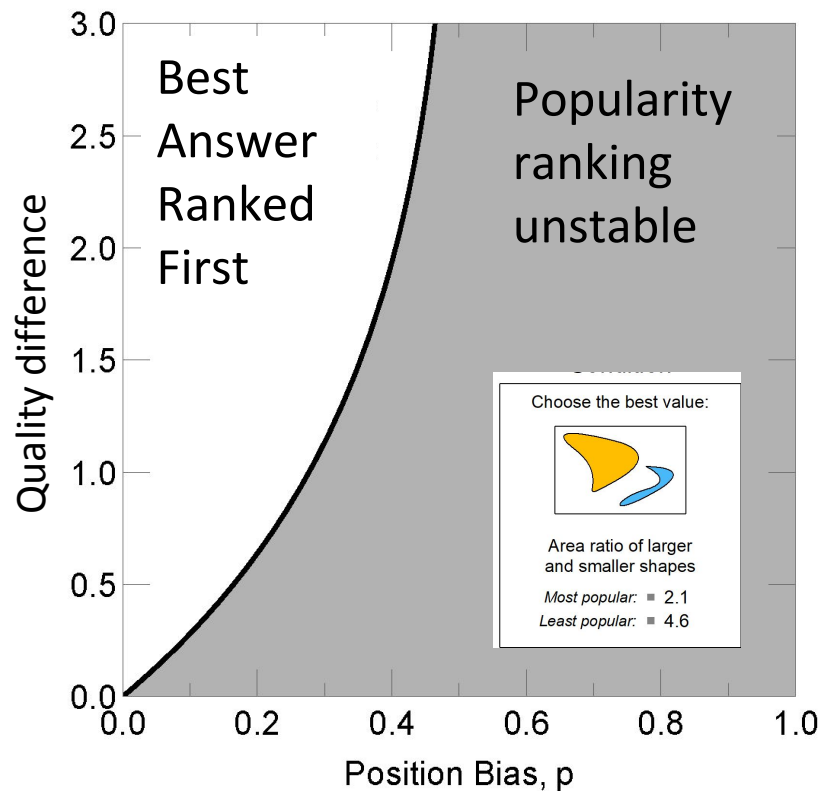
Good agreement with 2-parameter model





Algorithmic ranking instability

Burghardt et al. (2020) “Origins of algorithmic instabilities in crowdsourced rankings”, in *CSCW*.



- **Simulate a variety of cases: does correct answer float to the top?**
 - Items of different quality
 - Different position bias
- **Ranking instability: the objectively better answer does not always rise to the top, especially when**
 - Position bias too high (e.g., due to social influence)
 - Best answer is not unambiguously better
- Explain MusicLab results



Replicated findings experimentally

Finding The Best Answer To Questions

In the following pages, please find the most correct answer to each of ten questions on English language usage. We may ask you to explain why you chose each answer at the end of the task.

Important information:

- You can only perform this job once.
- Once you finish the job, a survey code will be shown.

Start

Please choose the best answer to the question from the list below.

What is the difference between 'nope' and 'no'?

What is the difference caused by using 'nope' instead of 'no'?

Is it used because "nope" sounds better and not straight like "no"? In some situations, it feels like nope is better to use than no even though it adds 2 more characters. What is the reason why many use it?

77

☐ 'Nope' is informal. You wouldn't usually want to use it, for example, answering questions for a job interview, but it would be an appropriate answer to a friend's question 'Have you seen that movie yet?'

45

☒ From [here](#), they have no difference in meaning; but nope is more informal, only used in a sense of opposite to yes (or yup). Also, nope is not used often in writing.

You wouldn't say 'there were nope errors', for example.

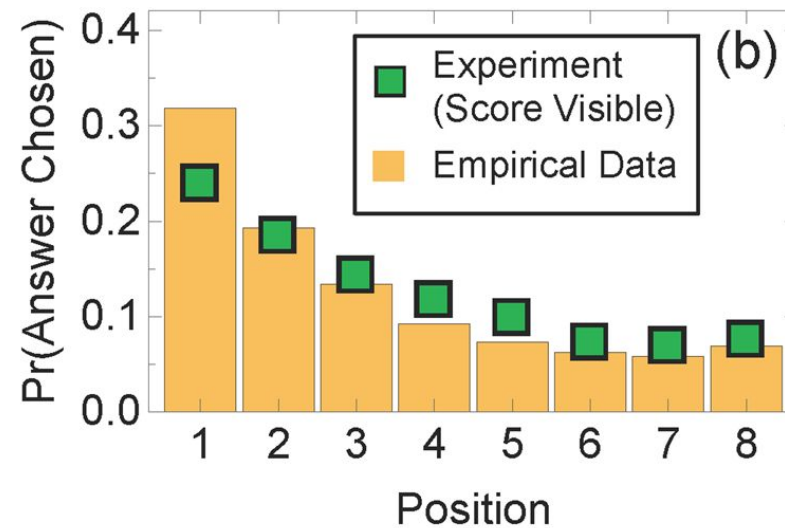
Next Question (9 Questions Left)

Turkers vote for best of 2 or 8 answers for each question



Strong position bias

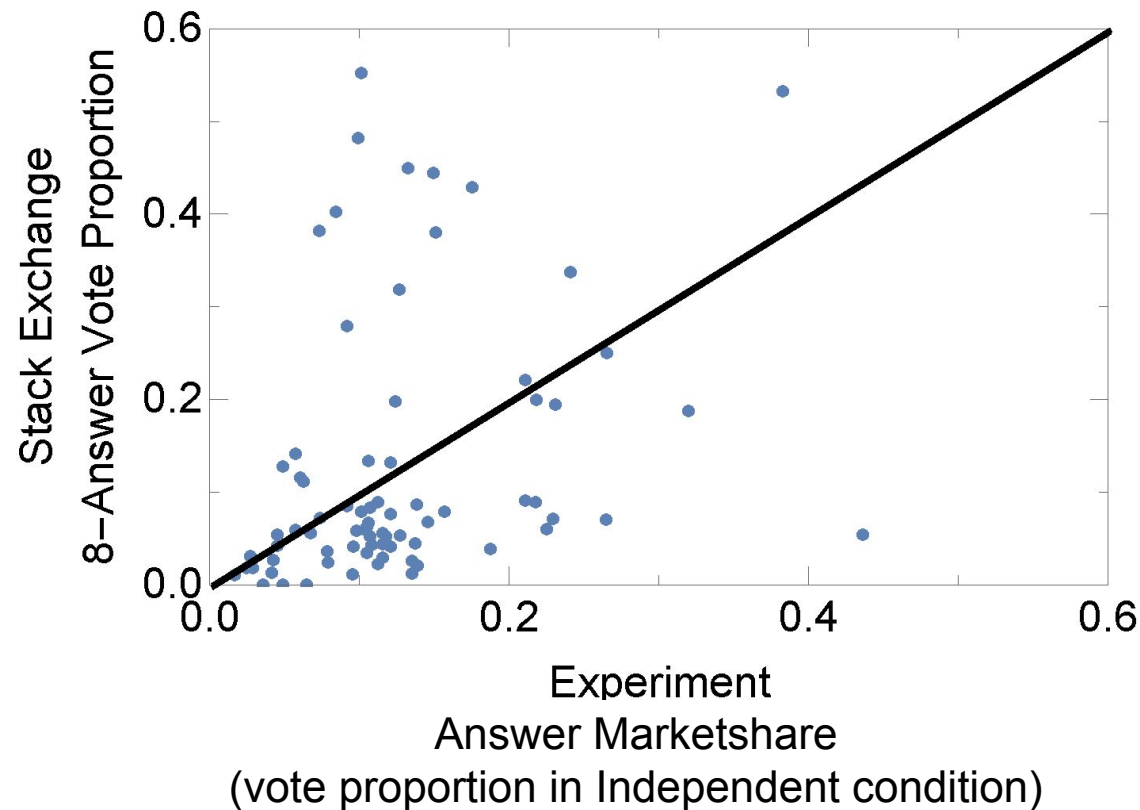
Probability to choose an answer in the experiment agrees with Stack Exchange data





Crowds do not identify best answers

Interplay with cognitive & alg. biases destabilize crowd wisdom



Insights



- **Understanding** the relationship between
 - Item quality
 - Item presentationand decision accuracy
- Quality helps explain algorithmic ranking instability
- Implications:
 - Crowdsourcing is messy (roughly **20% randomly choose answer**)
 - Random votes & position bias help create ranking instability
 - *Designers must reduce/control position bias* especially with the first few votes to improve crowdsourcing



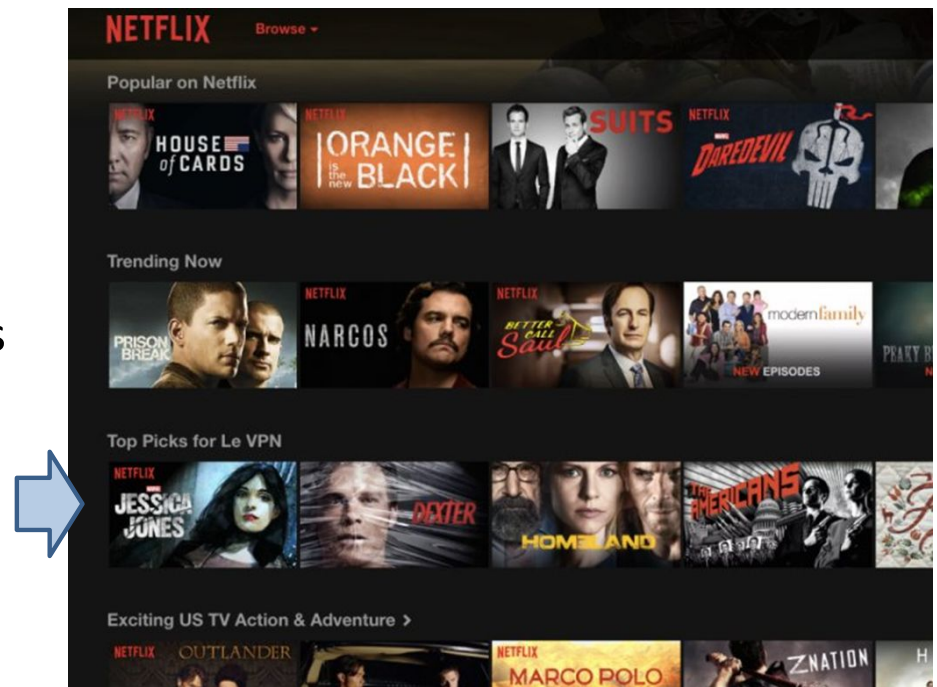
INSTABILITIES IN PERSONALIZATION



Instabilities in algorithmic recommendation

- Recommender systems:
 - Learn individual preferences to personalize content
 - Iteratively updates model of users based on feedback
- Model recommendation system to understand how user feedback affects algorithm performance
 - Agents with individual preferences
 - Agents act on recommended items
 - Algorithm updates
 - Which items become popular?

Personalized recommendation: Top picks for You





Recommender systems

- Personalization tools allow filtering large collections of movies, music, tv shows, ... to recommend only relevant items to people
 - Build a taste profile for a user
 - Build topic profile for an item
 - Recommend items that fit user's taste profile
- Probabilistic modeling techniques
 - Model people instead of documents to learn their profiles from observed actions
- Commercially successful (Netflix competition)

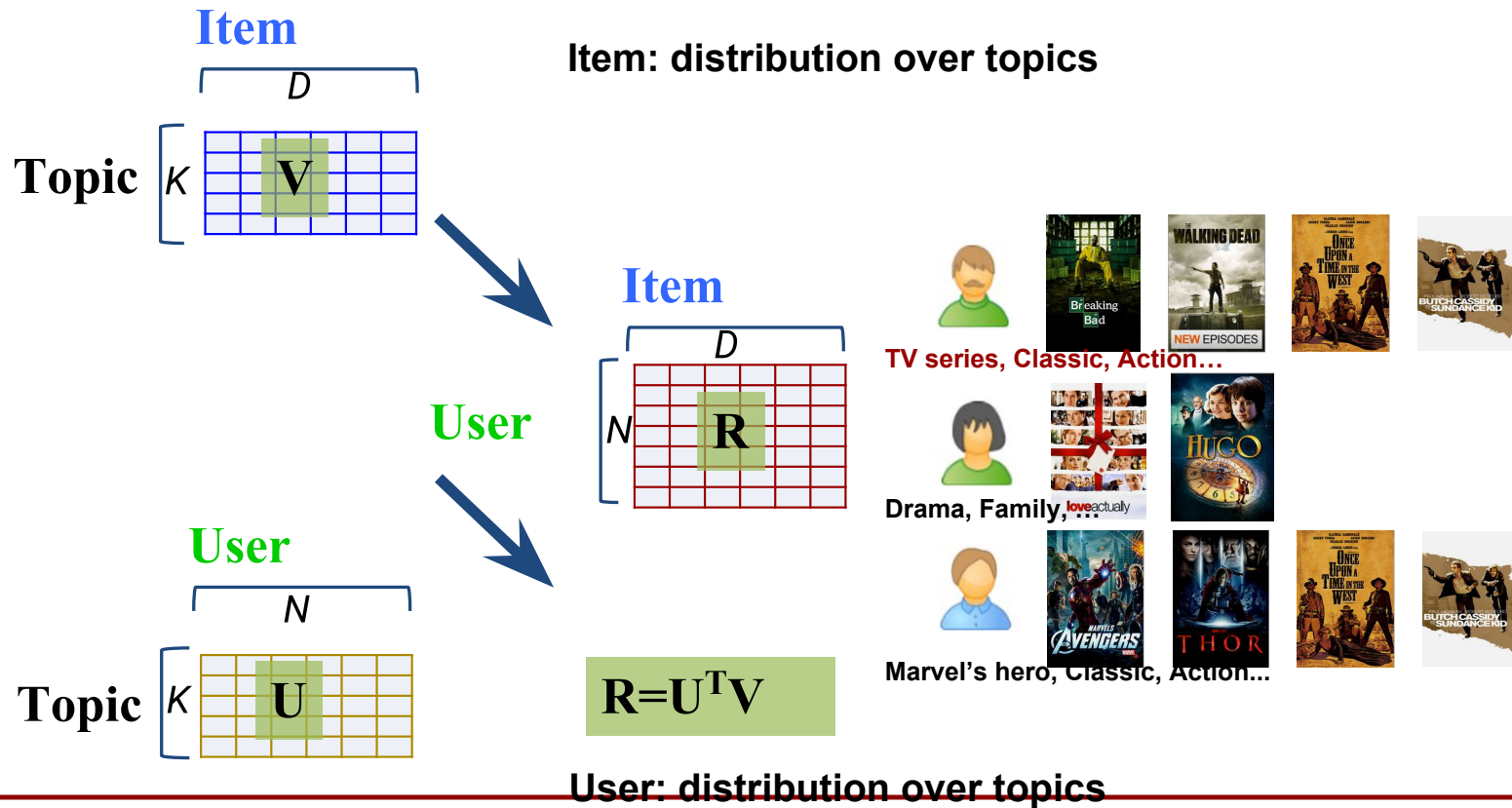


User-item rating prediction

		Items									
											...
Users		★★★★★	★★★★★ 5.0	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
		★★★★★	?	★★★★★	★★★★★	★★★★★	★★★★★		★★★★★	★★★★★	★★★★★
		?	★★★★★	★★★★★	★★★★★		★★★★★	★★★★★	★★★★★	★★★★★	
				★★★★★		?				?	
				★★★★★			?		?		
	...	★★★★★		?			?			★★★★★	
			?		★★★★★						?



Matrix Factorization





Application: Netflix challenge

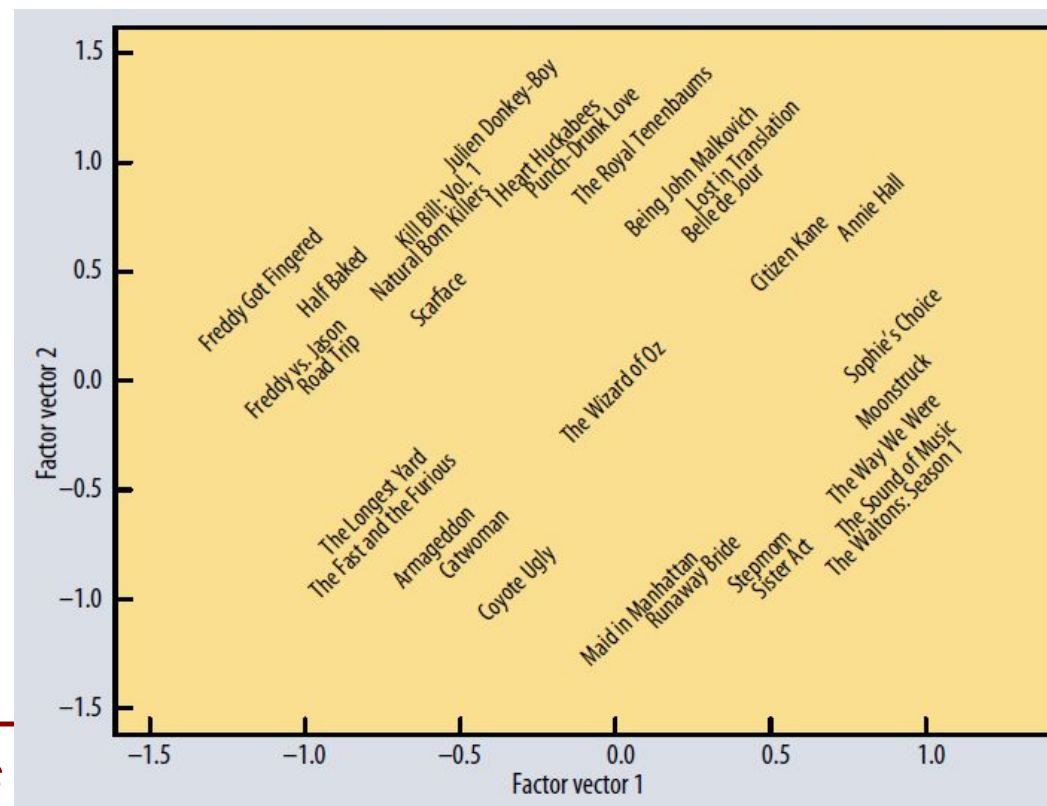
2006 contest to improve movie recommendations

- Data
 - 500K Netflix users (anonymized)
 - 17K movies
 - 100M ratings on scale of 1-5 stars
- Evaluation
 - Test set of 3M ratings (ground truth labels withheld)
 - Root-mean-square error (RMSE) on the test set
- Prize
 - \$1M for beating Netflix algorithm by 10% on RMSE
 - If no winner, \$50K prize to leading team



Factorization models in the Netflix competition

- Factorization models gave leading teams an advantage
 - Discover most descriptive “dimensions” for predicting movie preferences ...

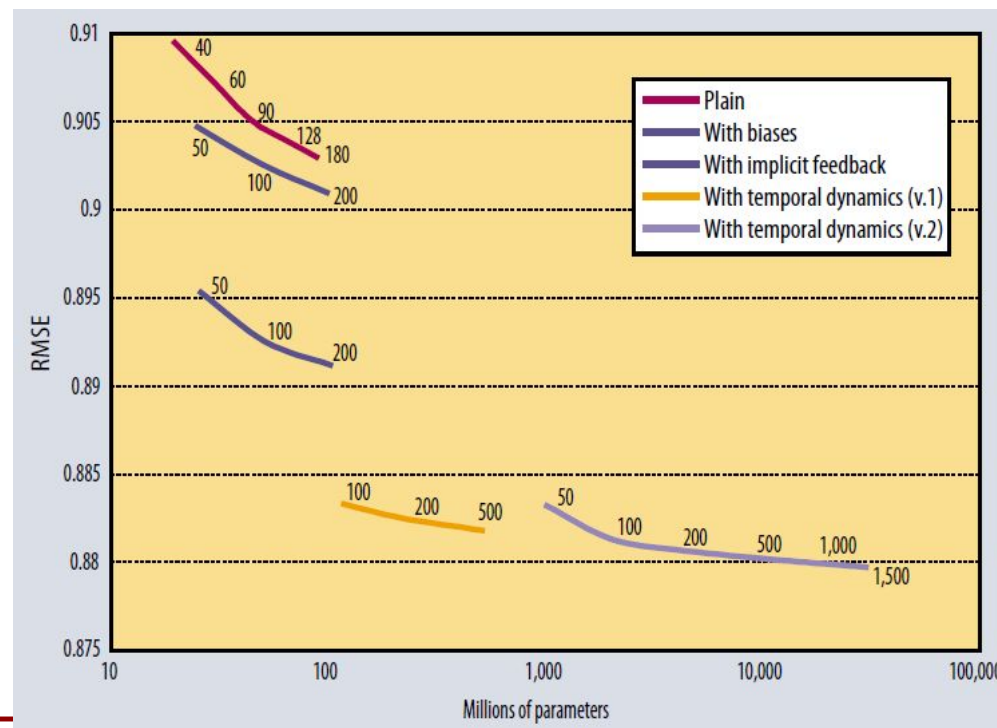




Performance of factorization models

- Model performance depends on complexity

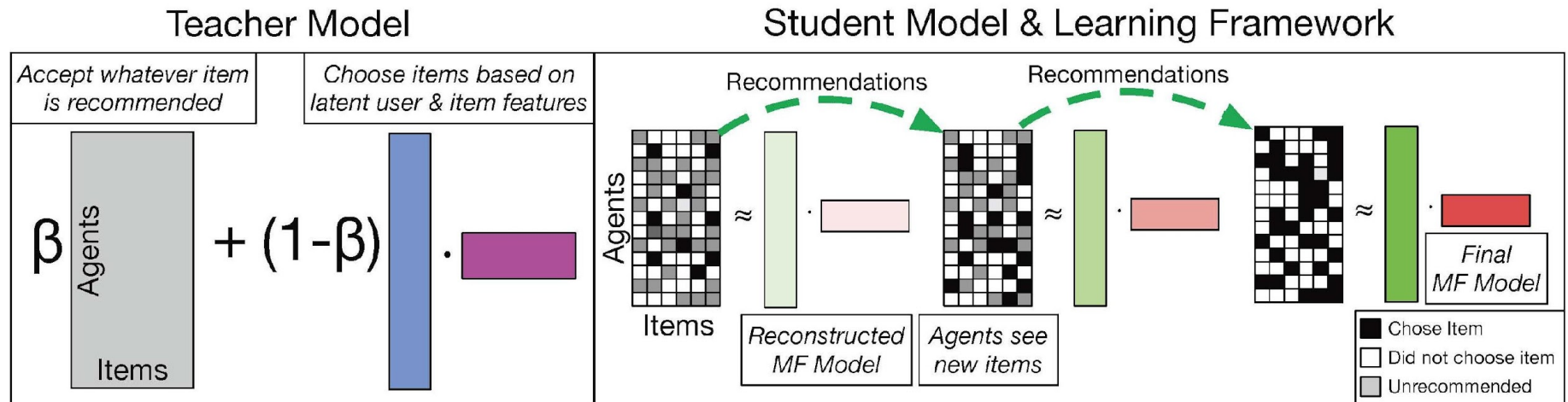
Netflix algorithm: RMSE=0.9514



Information Sciences Institute Grand prize target: RMSE=0.8563



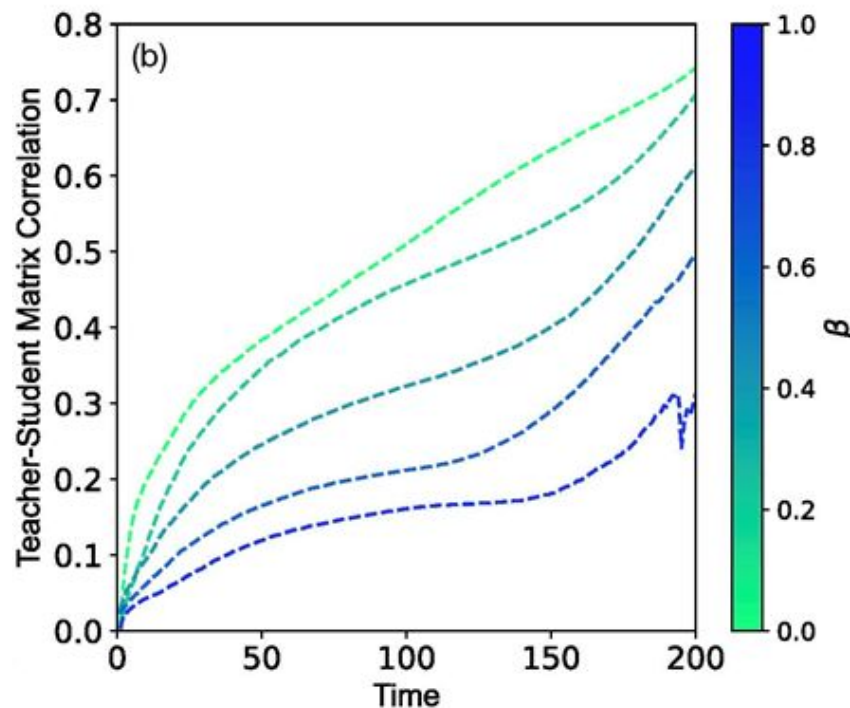
Model of a recommendation system



- Agents with **preferences**; items with **likabilities**
- Agents act on recommendations:
 - accept a recommended item with probability β ; otherwise, accept it according to preferences
- Column sum of agent $\times$ item matrix gives item popularity
- Recommender system learns agents' preferences using matrix factorization (MF)
- Agents' actions become new data to retrain the recommender system
- Popularity (how many agents chose the item): Column sum of agent $\times$ item matrix at time t



Learning is imperfect



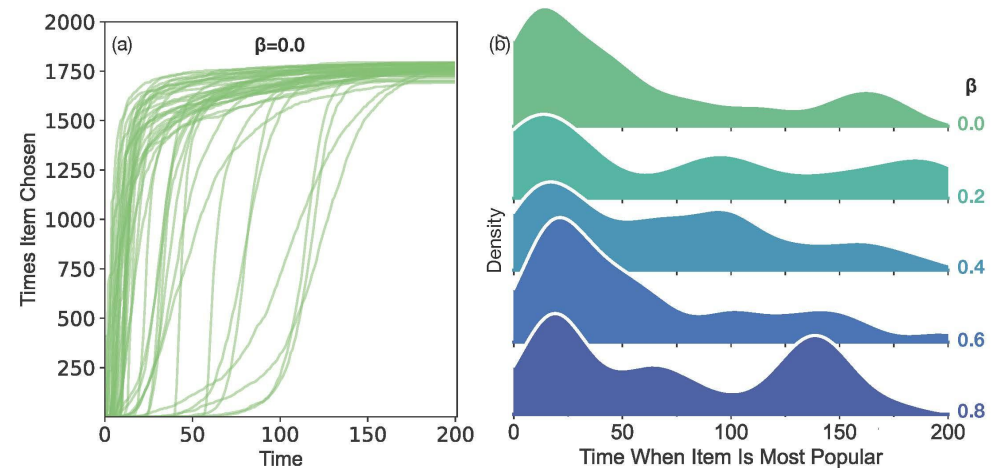
- Correlation between the teacher and student matrix elements
- More training data improves recommendations, but system can't learn ground truth (teacher model)
 - Learning is harder when user choice only partly reveals preferences (correlation drops with β)



Instability of the top item

- The best item eventually becomes the most popular, but ...
 - the time when it emerges as the most popular item varies significantly
- Recommendation is unstable
 - Algorithm is not guaranteed to find the top item until all data is seen
 - Slight change in initial conditions leads to large change in outcomes (popularity, time)

Item popularity over time, and distribution of times the top item emerges as most popular.

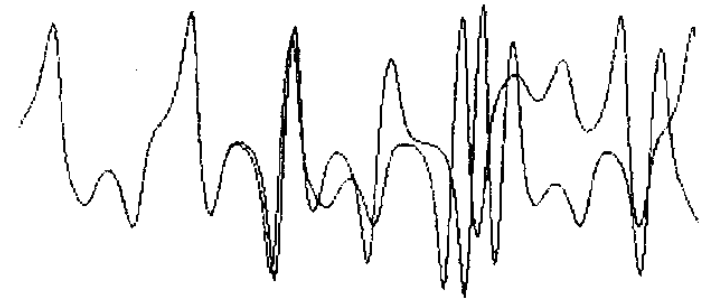


Algorithmic chaos

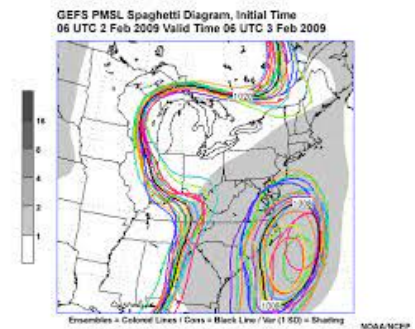


- Sensitivity to initial conditions ...
 - A small change in initial conditions can drastically change the long-term behavior of a system
 - The “butterfly effect” ☹
- ... is a hallmark of **chaos**
 - Makes even deterministic systems unpredictable

The “butterfly effect”



and the unpredictability of weather





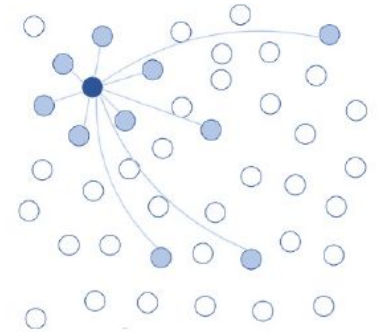
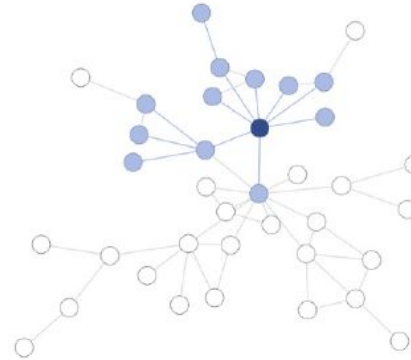
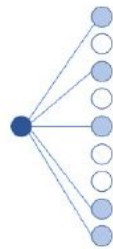
RECOMMENDER BIAS in SOCIAL MEDIA



Content recommendation

- Content recommendation/personalization algorithms are everywhere
 - Twitter feeds
 - YouTube sidebar recommendations
 - Netflix movie recommendation
 - Product recommendations
- Platforms using network models also use recommendation algorithms to rank a user's feed
 - Most users do not consume the entire feed; therefore, recommendation drives attention
- The core of recommendation is engagement prediction

Algorithms increasingly mediate social interactions



	Broadcast/Subscription model	Network model	Algorithm model
What a user sees	Posts by those you subscribed to	Posts by (or shared by) those you subscribed to	Posts the algorithm predicts the user will engage with
Examples	Substack, FB pre-2009, IG pre-2022	Word of mouth, the web, Twitter pre-2016	Tiktok, YouTube
What impacts reach	Poster's subscribed count	Subscribed count and content	Post's content

Source: A. Narayanan, UNDERSTANDING SOCIAL MEDIA RECOMMENDATION ALGORITHMS



Side effects of content recommendation

- Shift to recommendation reduces creators' control and predictability over content reach
 - Success highly volatile and dependent on viral posts
 - Virality is inherently unpredictable
- Recommendation algorithms optimize predicted engagement
 - Engagement optimization promotes impulsive (System 1) not deliberate (System 2) thinking.
 - Content that is provocative, polarizing, or emotionally charged tends to dominate, often undermining healthier discourse.
- Algorithmic neutrality and its failures
 - Platforms strive for algorithmic neutrality, avoiding explicit normative choices in content recommendation.
 - However, neutrality inadvertently amplifies pre-existing social biases and "rich-get-richer" phenomena, reinforcing inequality.

Source: A. Narayanan, UNDERSTANDING SOCIAL MEDIA RECOMMENDATION ALGORITHMS,



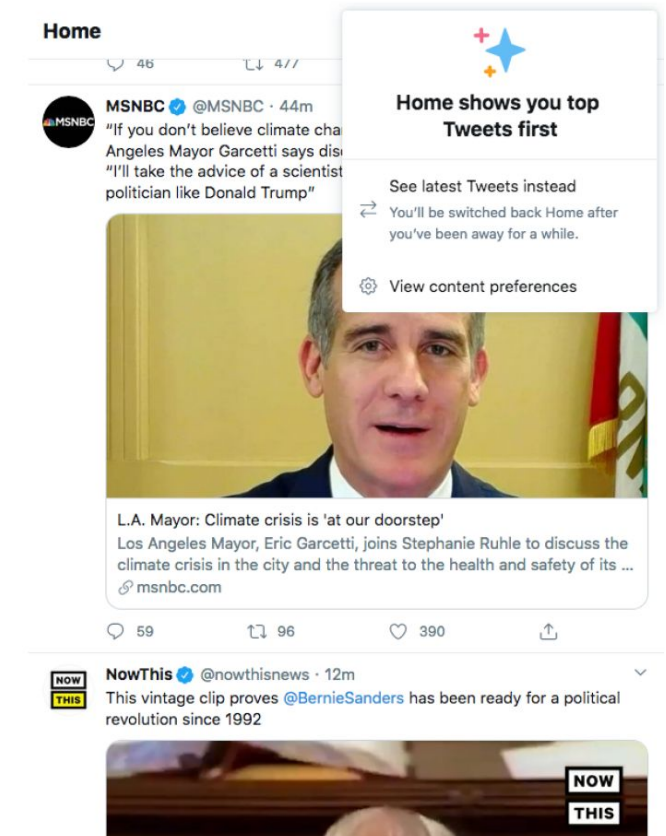
Different engagements to optimize

- Facebook optimizes for “Meaningful Social Interactions,” a weighted average of Likes, Reactions, Reshares, and Comments.
 - Initial MSI implementation unintentionally amplified divisive content due to high weights on reshares and comments.
 - Attempts at correction often backfire due to complex and nonlinear human-algorithm interactions.
- Twitter combines all the types of interaction that a user might have with a tweet.
 - Chronological feeds is skewed by frequent posters and viral content (is this bad?)
 - Information overload necessitates some form of ranking, thus simply reverting to chronological feeds is not a sustainable solution
- YouTube optimizes for expected watch time
 - how long the algorithm predicts the video will be watched. Prior to 2012, YouTube optimized for click-through rate, which led to clickbait thumbnails becoming ubiquitous
- Netflix optimized for suggesting movies that the user will like (rate high)



Auditing algorithmic bias on Twitter

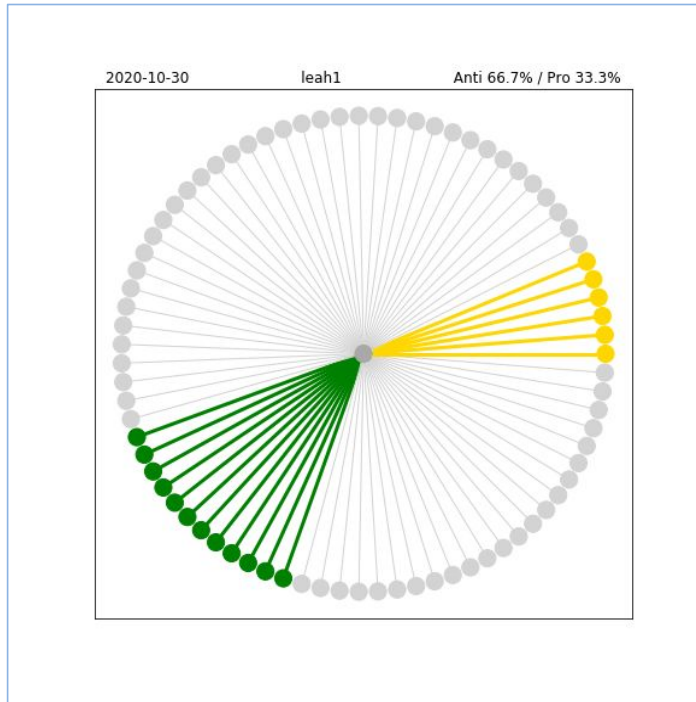
- Algorithmic exposure vs network exposure
- Does recommendation bias exposure?
 - A methodology to audit recommendation algorithms
 - Created synthetic users who follow same friends
 - They are passive: do not like, RT tweets
 - Differ in how they choose to see their timeline
 - Compare **personalized** feed (top tweets) to reverse **chronological** feed (latest tweets) to friends' **actual tweets**



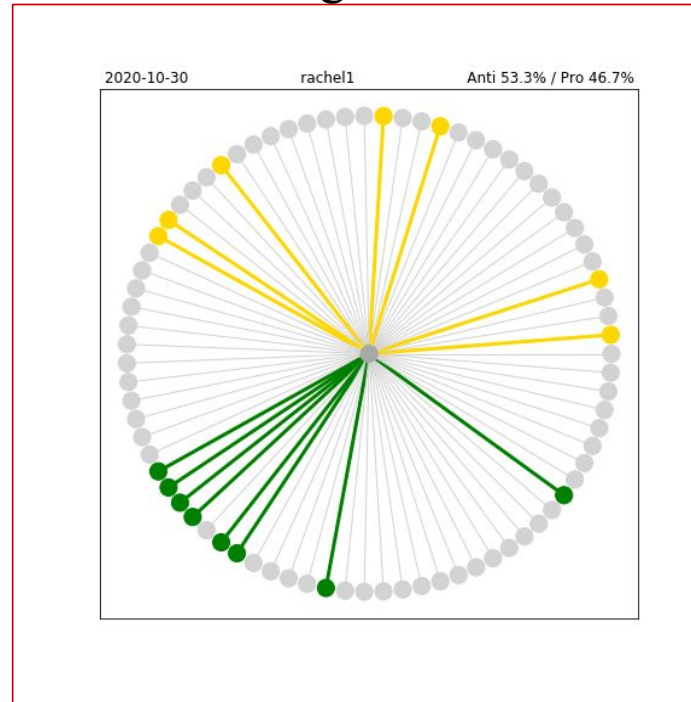


Personalization affects who is seen

A: Personalized feed



B: Chronological feed



- Pro-science accounts followed
- Anti-science accounts followed

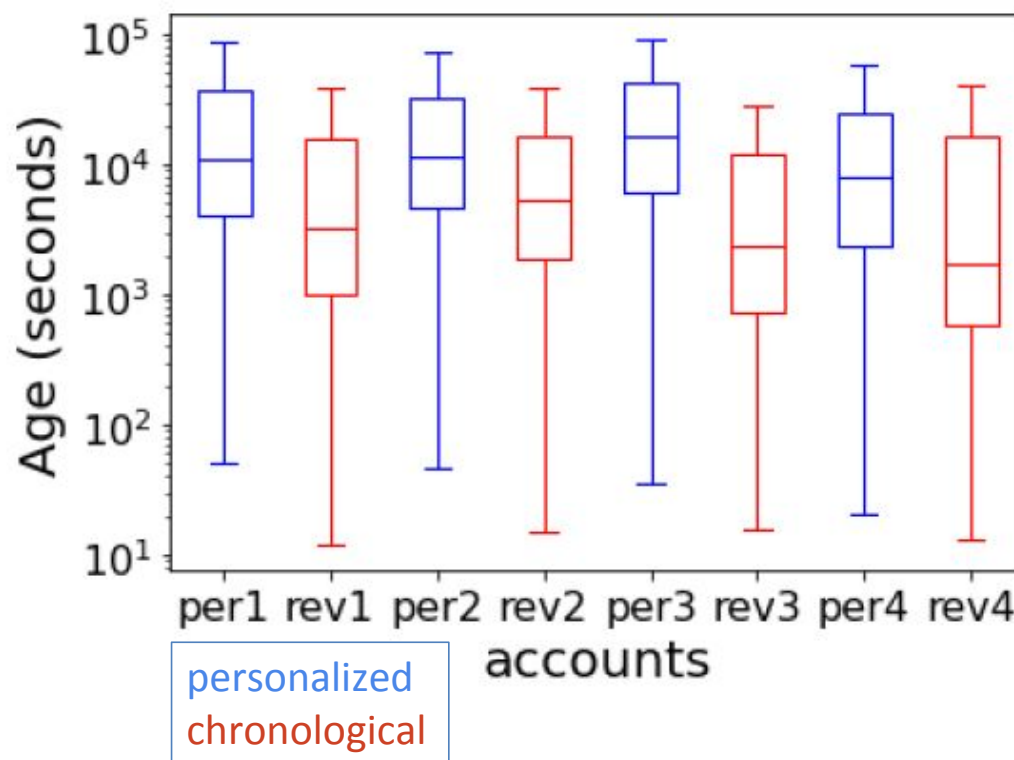
A and B follow the same accounts, log in at the same time, but due to personalization, A sees a tweets from only a subset of accounts A follows.

Recency Bias

Blue - personalized

Red - reverse chronological

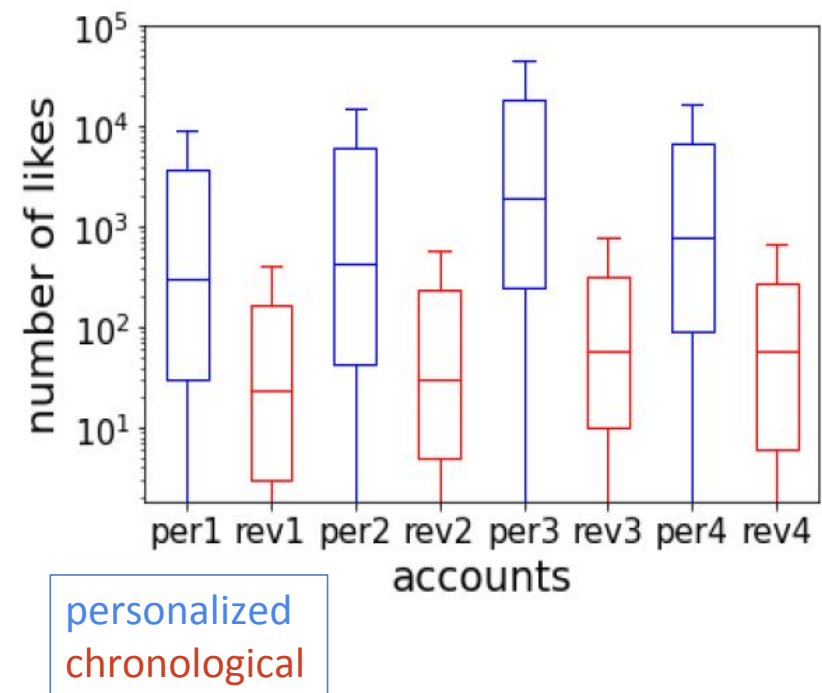
Personalized timelines tend to show **older** tweets





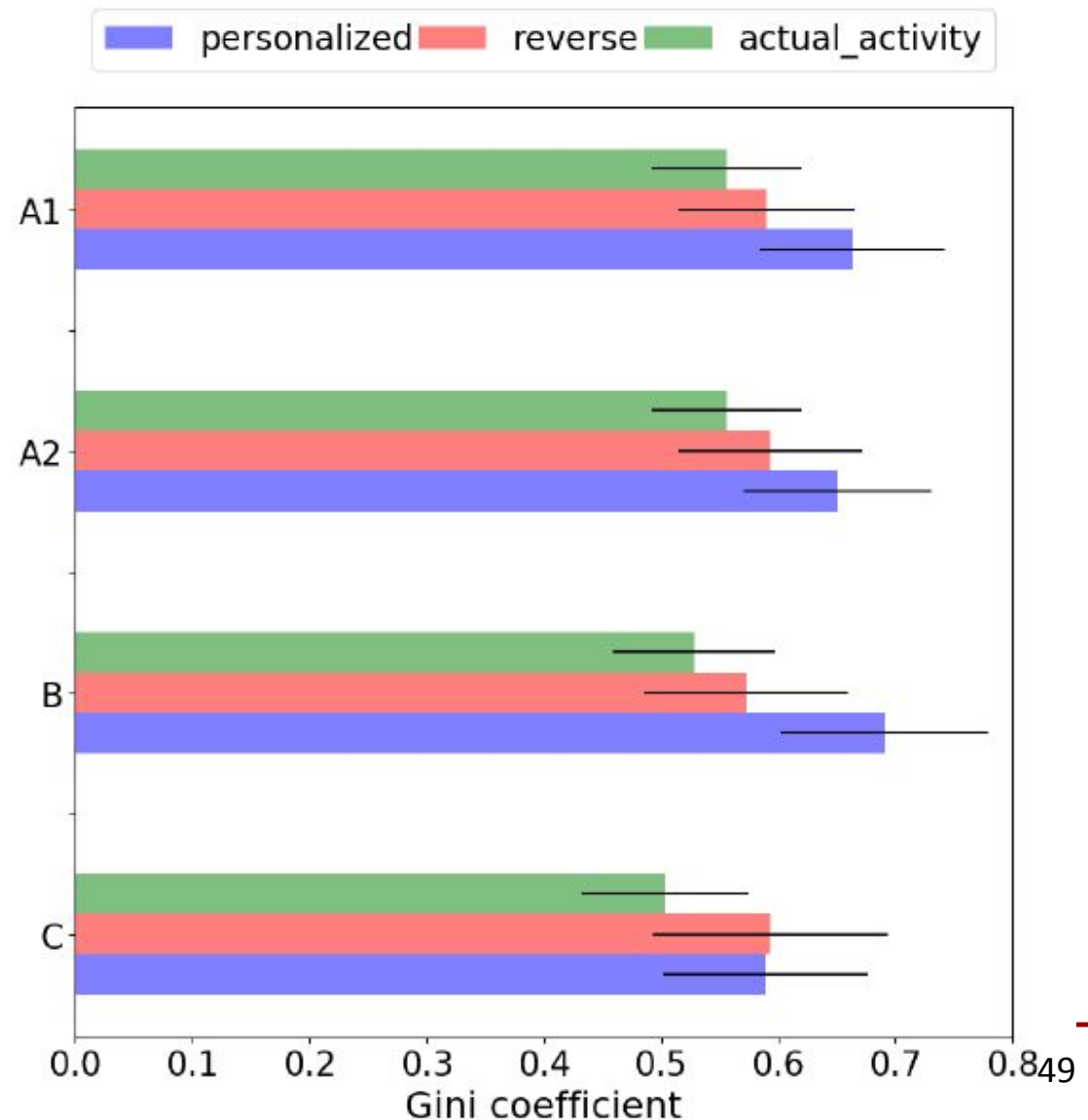
Popularity bias

Popularity bias: Personalized timelines show more **popular** tweets. Differences persist up to 2 months later



Exposure Bias

- Using Gini coefficient to measure inequality of exposure to friends
- Inequality already exists in friends' **activity** (some friends tweet much more than others)
- **Chronological** timeline **increases** exposure inequality
- **Personalization** **increases** inequality of exposure to friends, with a few friends shown more often in the timeline than others





RANKING INSTABILITIES IN GROWING NETWORKS



Ranking stability in complex networks

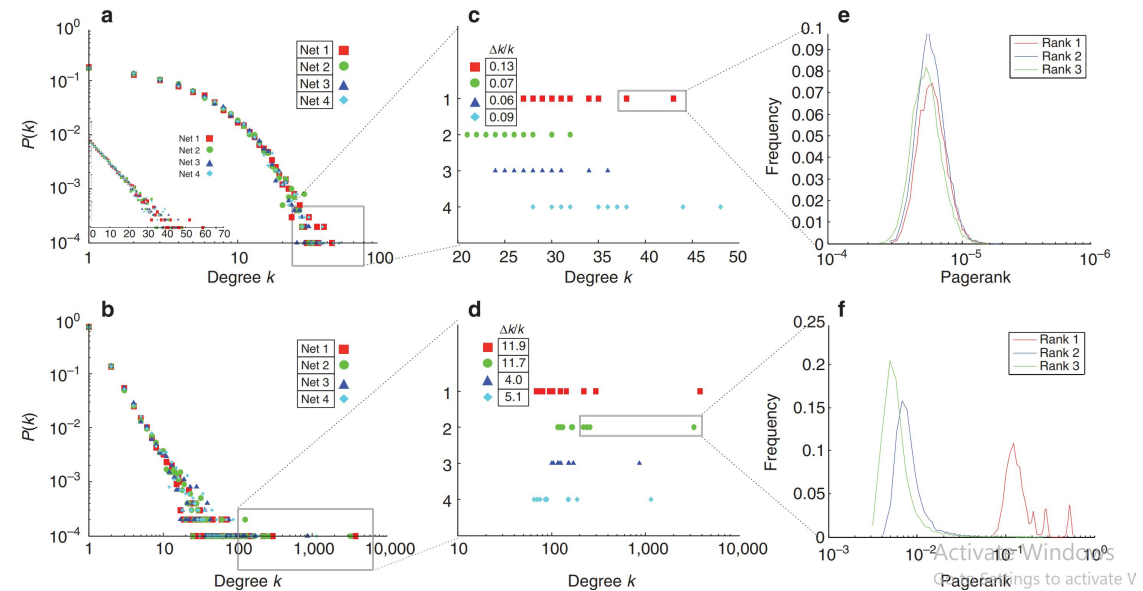
- How reliable is PageRank ranking?
- How does network structure impact ranking stability?
- Can nodes maintain their rank despite network perturbations and growth?
- **Methodology**
 - PageRank stability under network perturbations
 - Perturbations: random rewiring of network edges while preserving degrees
 - Growth: longitudinal data about growing networks (scientific citations, product purchasing)
 - Two main network types:
 - Scale-free networks (fat-tailed distributions)
 - Exponential networks (thin-tailed distributions)

Gourab Ghoshal & Albert-László Barabási (Nature Communications, 2011)



Emergence of super-stable nodes

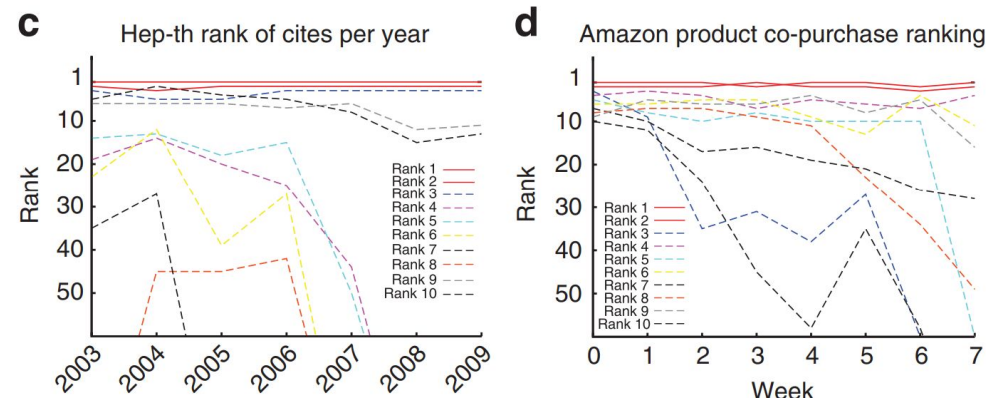
- "super-stable" nodes: nodes whose ranks remain stable despite perturbations
- Scale-free networks:
 - Super-stable nodes appear due to high-degree nodes (hubs)
 - Clearly separated top nodes
- Exponential networks:
 - No distinct super-stable nodes
 - Top nodes frequently change ranks





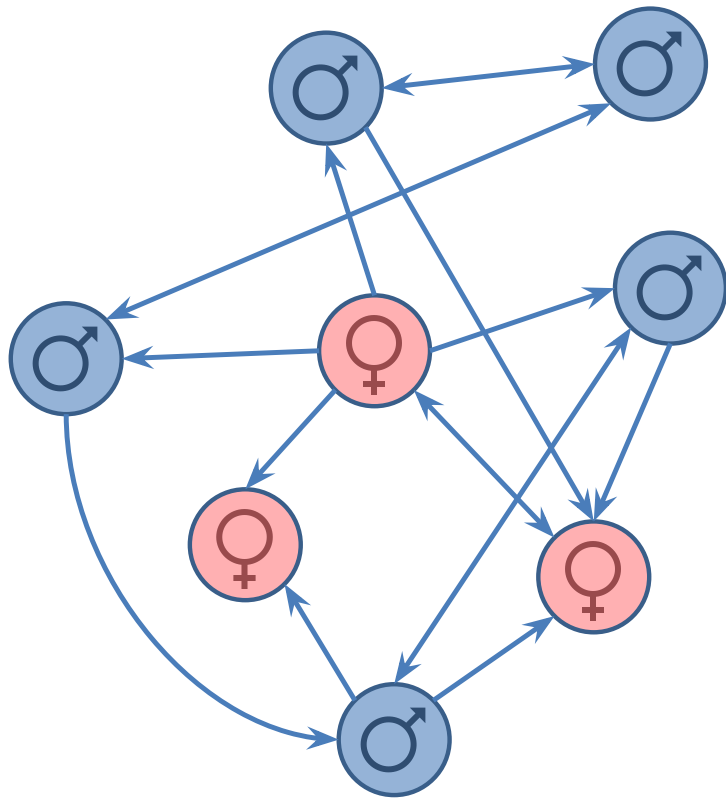
Temporal Stability of Super-Stable Nodes

- Identified super-stable nodes maintain rank over time
- Examples:
 - Highly-cited physics papers
 - Top-ranked Amazon products
- Non-super-stable nodes fluctuate widely in ranks over short periods
- Implications for fairness
 - Need fairness-aware ranking methods that counter structural biases
 - Highlight risk of cumulative advantage/disadvantage





Gender disparity in scientific citation networks



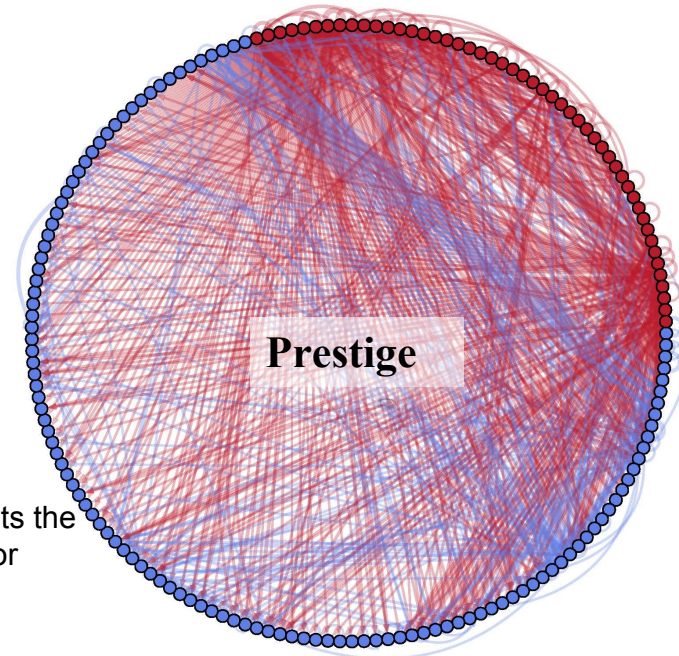
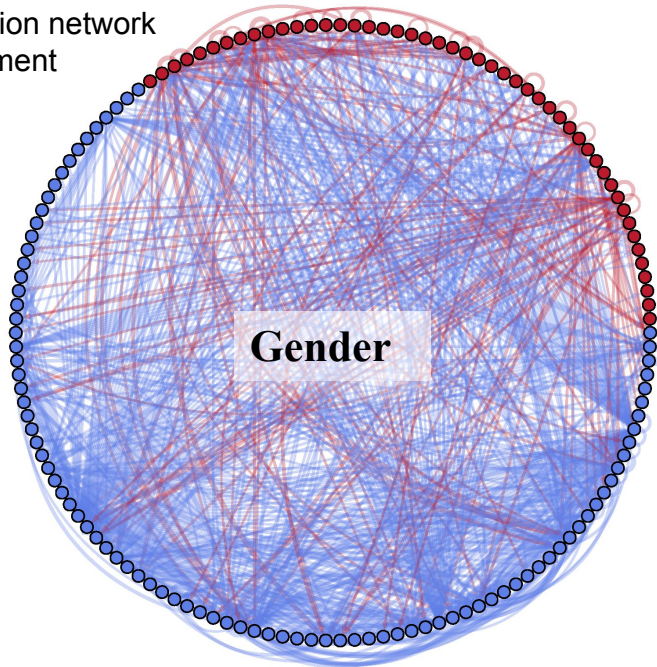
We study directed author citation networks

- A link $v \rightarrow u$ indicates that author u cites papers of author v
- Influence of v is $d_o(v)$
 - #authors who cited v
 - Out-degree of v
- Attention of v to others is $d_i(v)$
 - #authors that v cited
 - In-degree of v



Inequalities in Author-Citation Networks

Subgraph of
author-citation network
in Management



Each edge inherits the
color of the author
being cited

Female and

Male

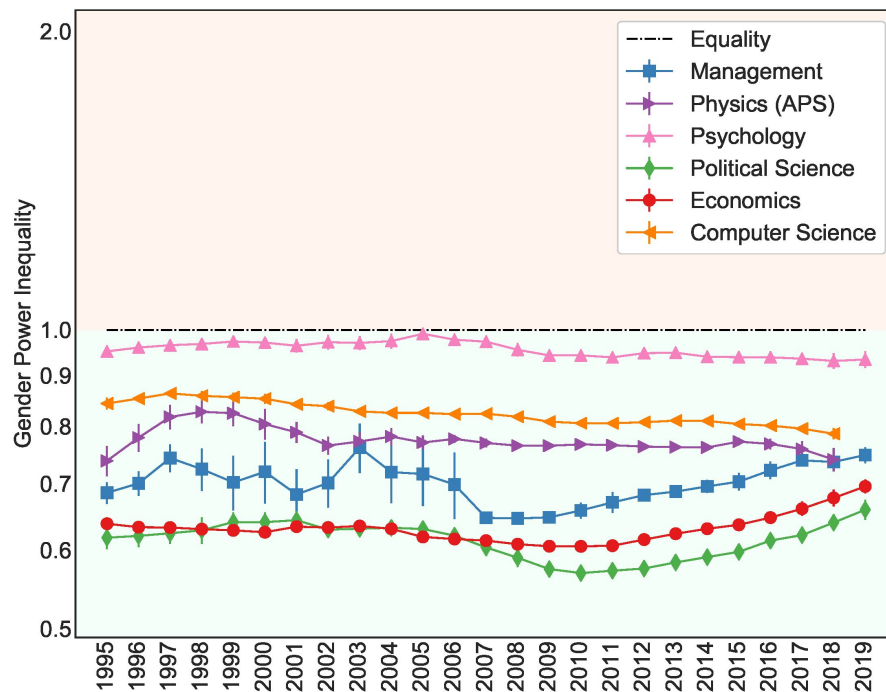
Elite (top-100) and

Other

- Size of the group does not explain the disparity
- How to quantify the observed disparity?



Gender bias in citations

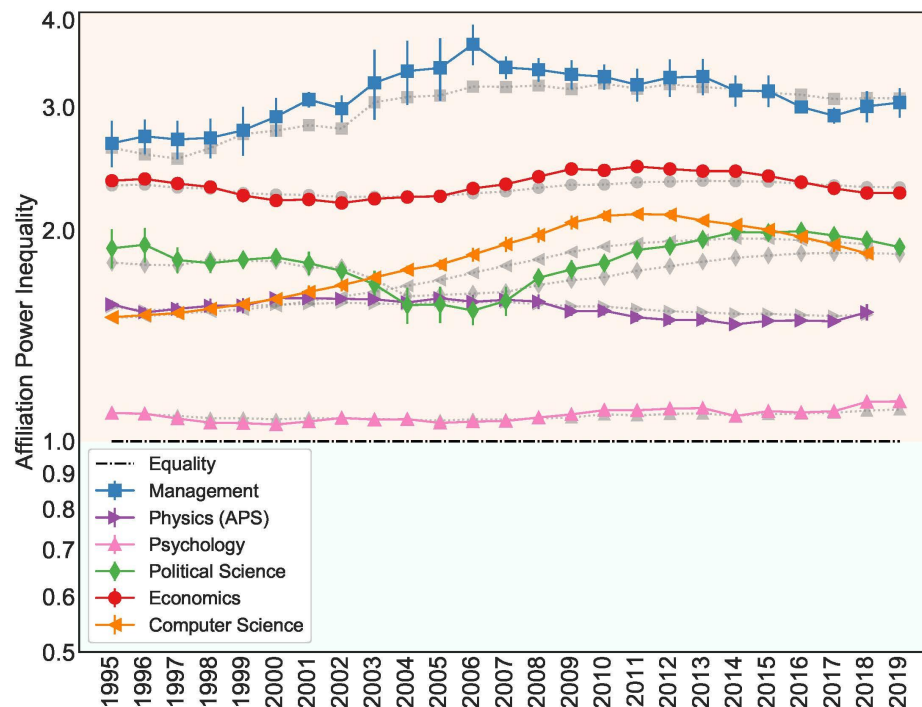


- Two groups
 - Minority (R): women
 - Majority (B): men
 - Power of group R is $d_o(R)/d_i(R)$
- Power inequality is
 - Power of Red group / power of Blue group
 - $I_t = d_o(R)/d_i(R) \cdot d_i(B)/d_o(B)$
- Power inequality < 1
 - Women have less power than men
- Most parity: Psychology
- Most disparity: Political Science, Economics

[Nettasinghe et al. (2021). Emergence of Structural Inequalities in Scientific Citation Networks. arXiv:2103.10944]



Prestige bias in citations

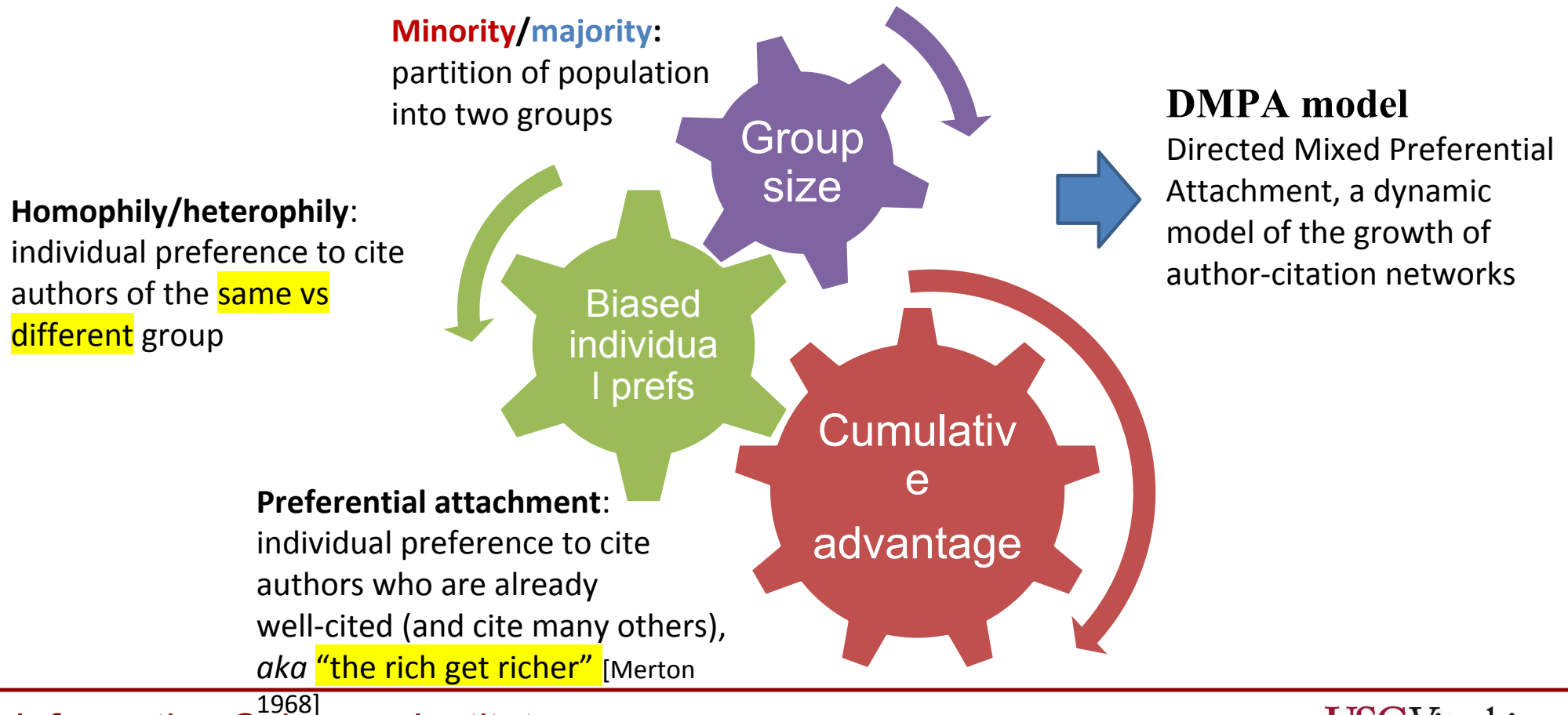


- Two groups
 - Minority (R): elite authors (top-100 affiliations)
 - Majority (B): authors with other affiliations
- Power inequality > 1
 - Elite authors have more power than others
- Most parity: Psychology
- Most disparity: Management, Economics

[Nettasinghe et al. (2021). Emergence of Structural Inequalities in Scientific Citation Networks. arXiv:2103.10944]



Emergence of gender disparity in citations

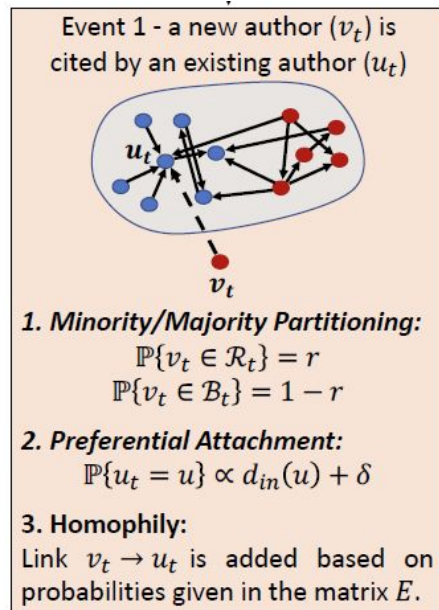




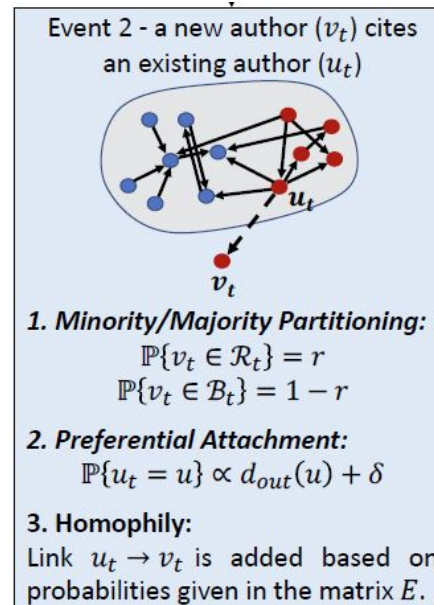
Emergence of gender disparity in citations

At each time step $t=1, 2, \dots$

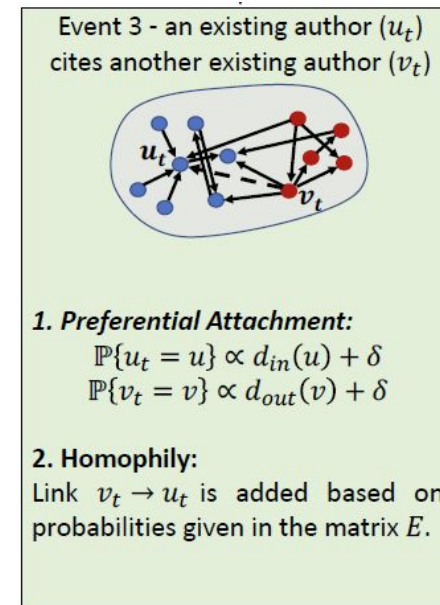
Probability p



Probability q



Probability $1 - p - q$



p, q give the rate at which new authors arrive

- p : new authors is cited by an existing author
- q : new author cites an existing author
- $1 - p - q$: existing authors cite each other

r determines the majority:

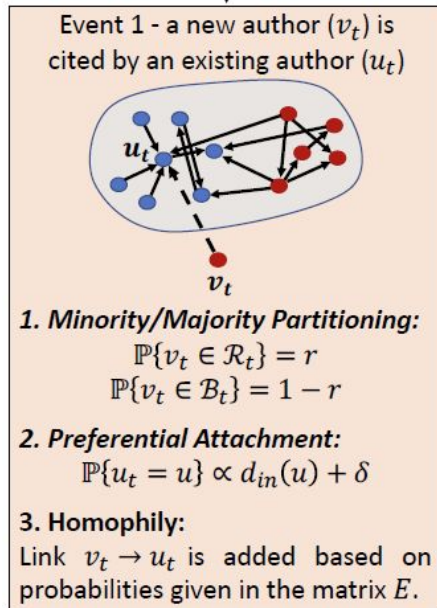
- $r > 0.5$ as $t \rightarrow \infty$, red group R will be the majority
- $r < 0.5$ as $t \rightarrow \infty$, blue group B will be the majority
- $r = 0.5$ as $t \rightarrow \infty$, both groups will be of equal sizes



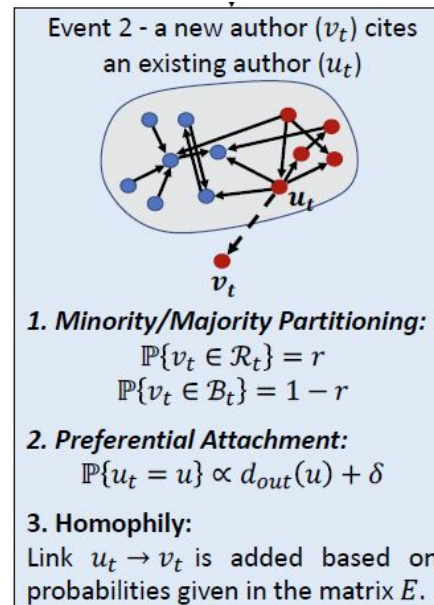
Emergence of gender disparity in citations

At each time step $t=1, 2, \dots$

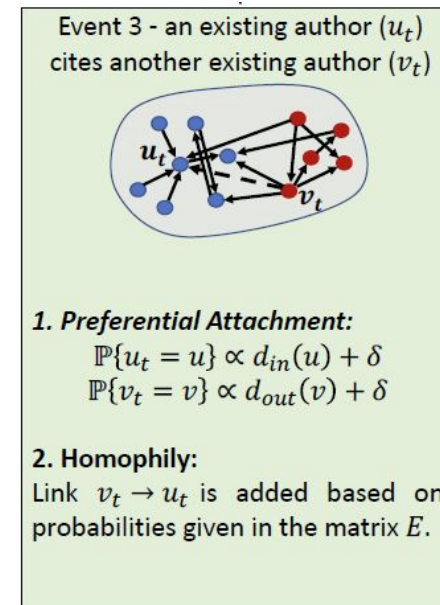
Probability p



Probability q



Probability $1 - p - q$



$\delta > 0$ determines the level of preferential attachment e.g.,

- $\delta \gg 1$, less preferential attachment in both citing and being cited
- $\delta \sim 1$, high preferential attachment in both citing and being cited

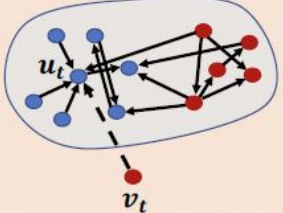


Emergence of gender disparity in citations

At each time step $t=1, 2, \dots$

Probability p

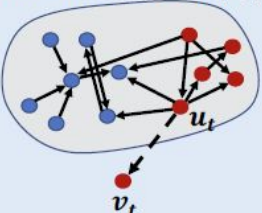
Event 1 - a new author (v_t) is cited by an existing author (u_t)



- Minority/Majority Partitioning:**
 $\mathbb{P}\{v_t \in \mathcal{R}_t\} = r$
 $\mathbb{P}\{v_t \in \mathcal{B}_t\} = 1 - r$
- Preferential Attachment:**
 $\mathbb{P}\{u_t = u\} \propto d_{in}(u) + \delta$
- Homophily:**
 Link $v_t \rightarrow u_t$ is added based on probabilities given in the matrix E .

Probability q

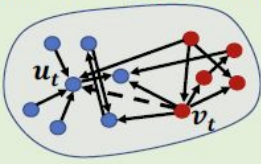
Event 2 - a new author (v_t) cites an existing author (u_t)



- Minority/Majority Partitioning:**
 $\mathbb{P}\{v_t \in \mathcal{R}_t\} = r$
 $\mathbb{P}\{v_t \in \mathcal{B}_t\} = 1 - r$
- Preferential Attachment:**
 $\mathbb{P}\{u_t = u\} \propto d_{out}(u) + \delta$
- Homophily:**
 Link $u_t \rightarrow v_t$ is added based on probabilities given in the matrix E .

Probability $1 - p - q$

Event 3 - an existing author (u_t) cites another existing author (v_t)



- Preferential Attachment:**
 $\mathbb{P}\{u_t = u\} \propto d_{in}(u) + \delta$
 $\mathbb{P}\{v_t = v\} \propto d_{out}(v) + \delta$
- Homophily:**
 Link $v_t \rightarrow u_t$ is added based on probabilities given in the matrix E .

E determines the level of homophily e.g.,

- $\rho_R > 0.5$, blue group is **homophilic** (more likely to cite other blue nodes)
- $\rho_A < 0.5$, blue group is **heterophilic** (more likely to cite red nodes)

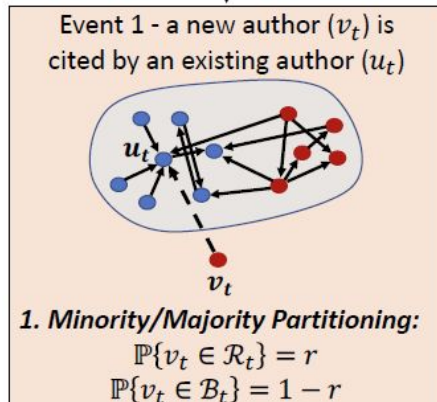
$$E = \begin{matrix} \mathcal{B} & \mathcal{R} \\ \mathcal{B} & \begin{bmatrix} \rho_B & 1 - \rho_R \\ 1 - \rho_B & \rho_R \end{bmatrix} \\ \mathcal{R} & \end{matrix}$$



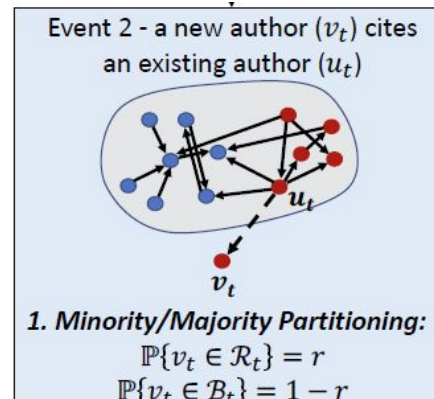
Emergence of gender disparity in citations

At each time step $t=1, 2, \dots$

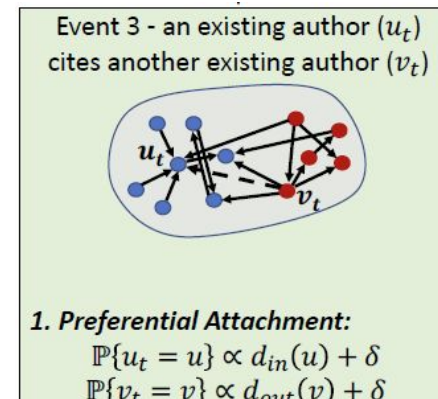
Probability p



Probability q



Probability $1 - p - q$



Model has a steady state solution!

- Closed-form expressions for the exponents of the asymptotic degree distributions of Red and Blue nodes [1]
- Analytic solution for the power inequality [2]

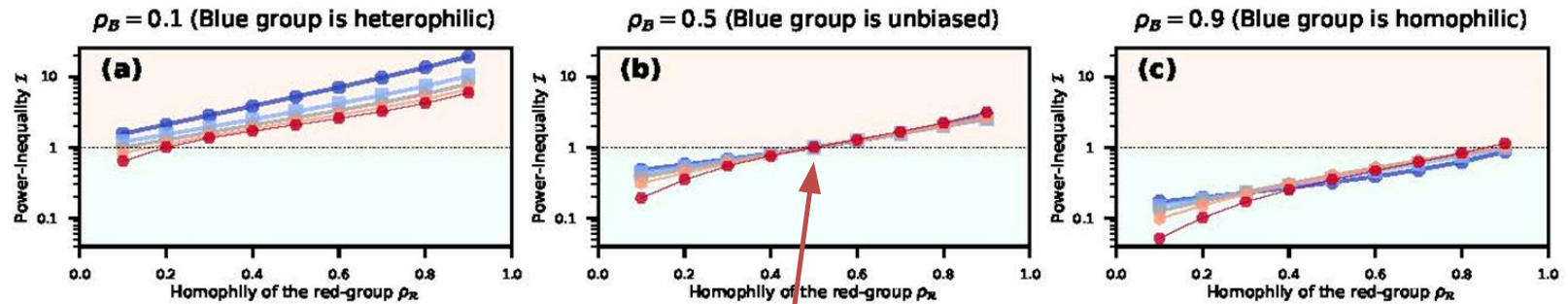
1. Nettasinghe et al (2022). Scale-free degree distributions, homophily and the glass ceiling effect in directed networks. *JCN* 10(2), cnac007.
2. Nettasinghe et al (2021). Emergence of structural inequalities in scientific citation networks.



Using the model to explore citations disparities

(Red is minority) $r=0.1$ $r=0.3$ $r=0.5$ $r=0.7$ $r=0.9$ (Red is majority)

New authors
are infrequent
 $p = 0.1$,
 $q = 0.2$



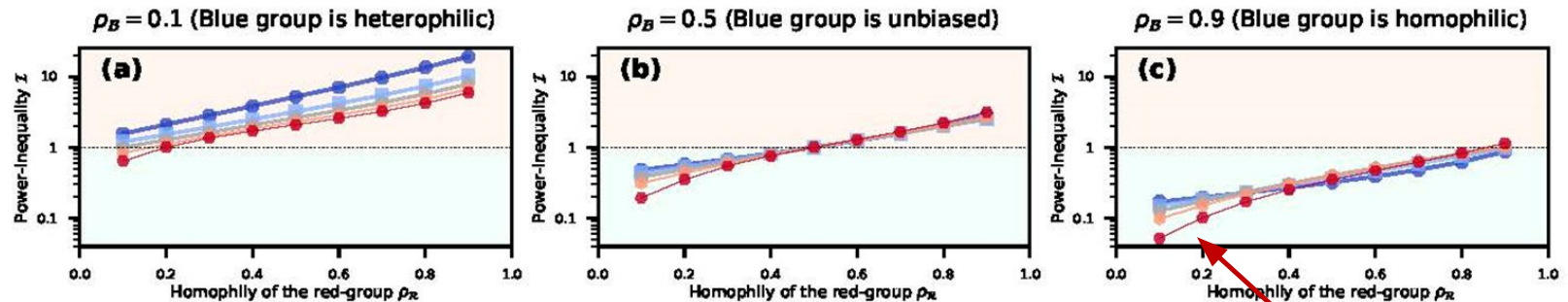
No power inequality when no
homophily ($\rho_B = \rho_A = 0.5$)
regardless of all other
parameters



Using the model to explore citations disparities

(Red is minority) $r = 0.1$ $r = 0.3$ $r = 0.5$ $r = 0.7$ $r = 0.9$ (Red is majority)

New authors
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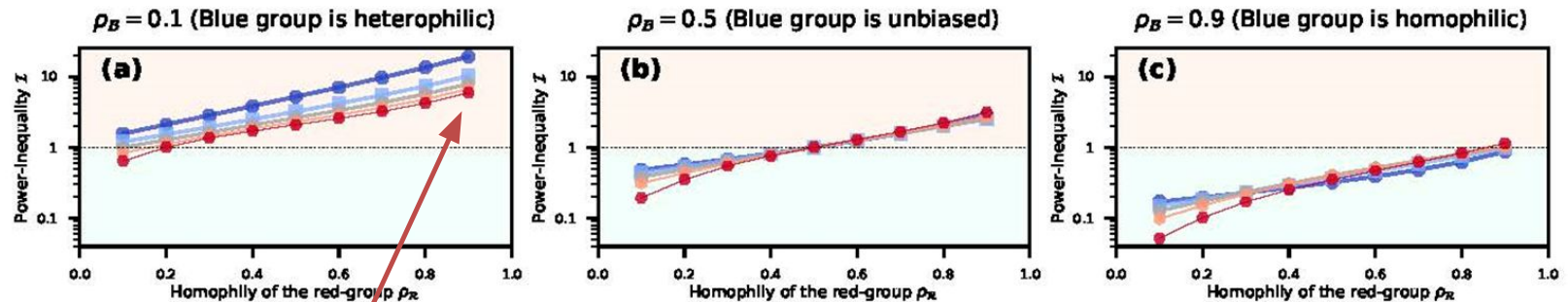
When blue group (men) is homophilic ($\rho_B \gg 0.5$) and red is heterophilic ($0.5 \gg \rho_A$), red (women) has less power and increasing its size will give even more power to the blue group (men)



Using the model to explore citations disparities

(Red is minority) $r=0.1$ $r=0.3$ $r=0.5$ $r=0.7$ $r=0.9$ (Red is majority)

New authors
are infrequent
 $p = 0.1$,
 $q = 0.2$



When red group (elites) is homophilic ($\rho_A \gg 0.5$) and blue heterophilic ($0.5 \gg \rho_B$), red (elites) has more power than blue (non-elites) and decreasing its size will give it even more power, e.g., top-25 instead of top-100 universities

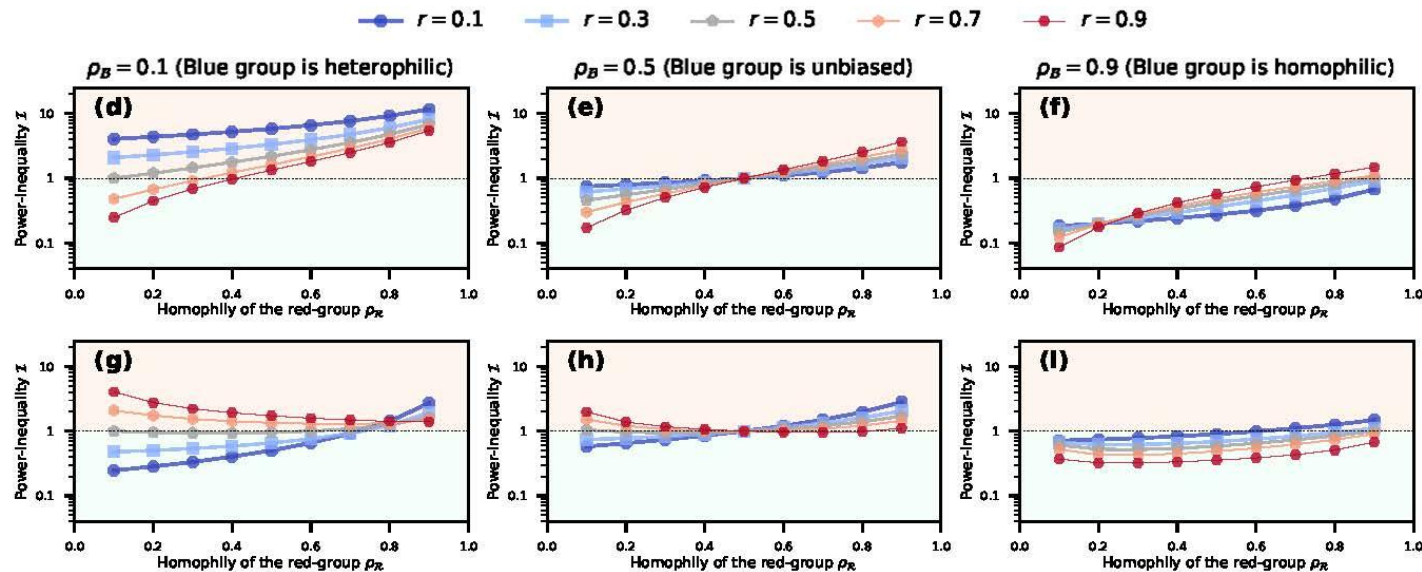


Using the model to explore citations disparities

New authors join frequently

$p = 0.1,$
 $q = 0.7$

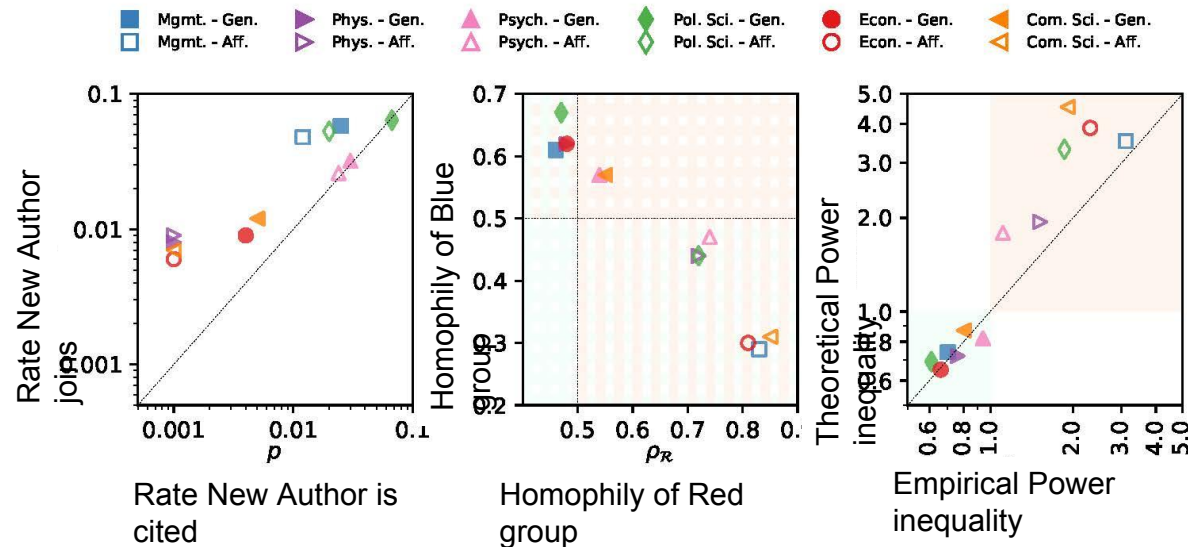
$p = 0.7,$
 $q = 0.1$



- Arrival of new authors reduces power inequality, esp when they are recognized by existing authors
- Reducing preferential attachment reduces inequality (increasing δ)



Estimating model parameters from bibliographic data



- Agreement between theoretical and empirical values of power disparity
- Worse gender disparity in citations when blue nodes (men) are homophilic and red (women) are heterophilic
- Prestige disparity is high when red nodes (elites) are homophilic and blue nodes are heterophilic

Insights



- Disparities in citations arise due to social mechanisms such as preference to cite similar (or dissimilar) others, and the reward mechanisms of science (cumulative advantage)
- These disadvantages accumulate [Rossiter 1993], creating barriers to women's career advancement
- It is difficult to reduce citations gender disparity simply by increasing the number of women
- Instead, need to change homophily and reduce preferential attachment

Summary



- **Trust and predictability**
- **Socially-embedded algorithms are inherently unpredictable**
 - Can we trust unpredictable systems?
 - Can we design human-centered AI systems with unpredictability in mind?